1 2

1 Home_Die_Motor

2 LatchHomeDie PolyPipe.Ax1In3

2.1

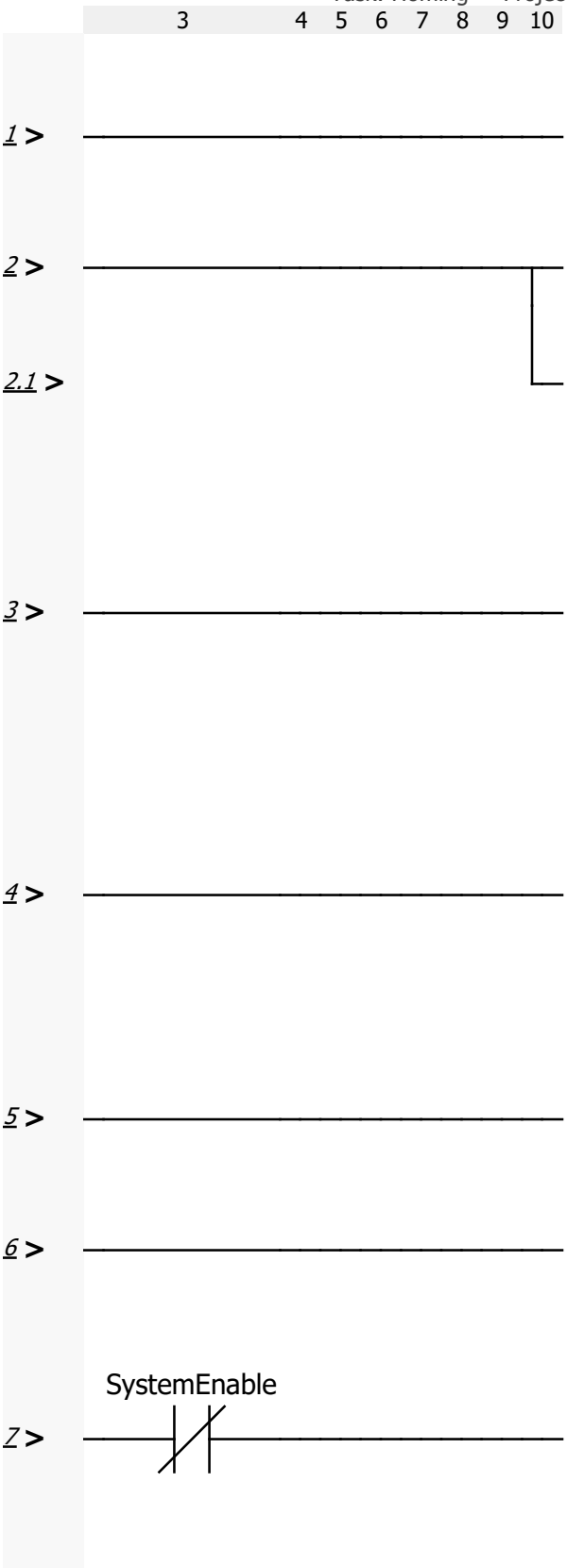
3 LatchHomeDie SystemEnable

4 LatchHomeDie SystemEnable

5 Home_Table_Motor

6 LatchHomeTable PotentiometerPositionMM < Potent_Edge_scale_mm

7 LatchHomeTable PotentiometerPositionMM > Potent_Edge_scale_mm


3 4 5 6 7 8 9 10
1 >
2 >
2.1 >
3 >
4 >
5 >
6 >
SystemEnable
7 >

1 > LatchHomeDie ( SET )

2 > LatchHomeDie ( RST )

2.1 > Enable ↑
SET POSITION
Module Name: AMC-1
Channel/Axis: 1 Channel/Axis Name: Die Motor
Set Position To 0

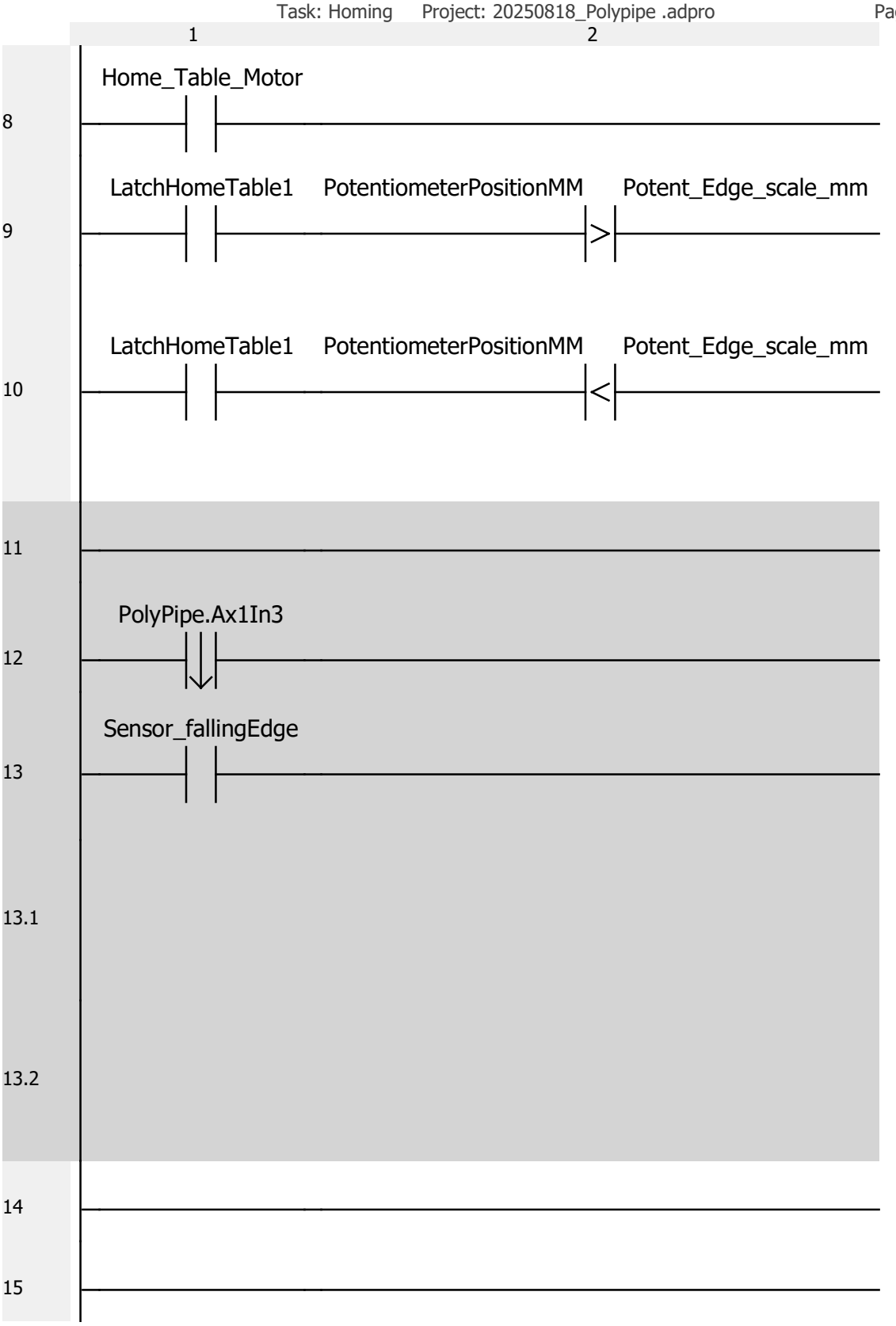
3 > Enable
VELOCITY MOVE
Module Name: AMC-1
Channel/Axis: 1 Channel/Axis Name: Die Motor
Move Direction Positive
Velocity 0.5
Ramp Rate 1
Stop Setup Immediate Stop

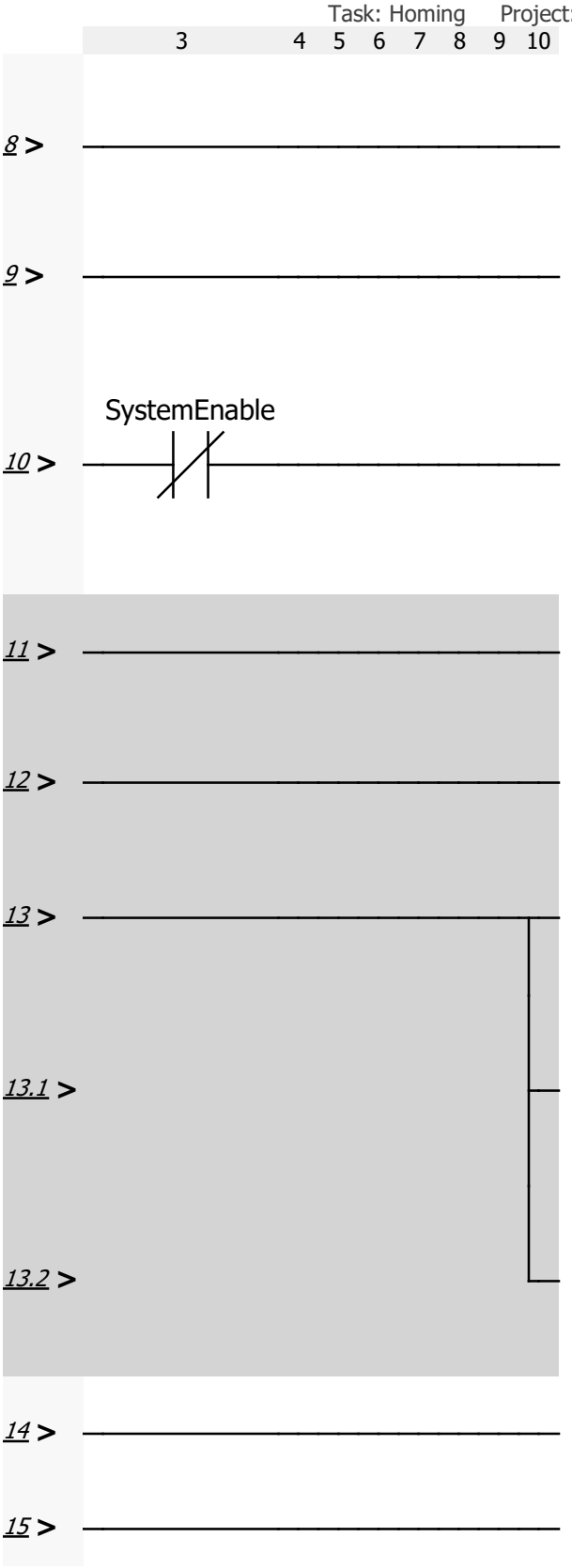
4 > Enable
VELOCITY MOVE
Module Name: AMC-1
Channel/Axis: 2 Channel/Axis Name: Top Motor
Move Direction Negative
Velocity 2
Ramp Rate 4
Stop Setup Immediate Stop

5 > LatchHomeTable ( SET )

6 > LatchHomeTable ( RST )

7 > Enable
VELOCITY MOVE
Module Name: AMC-1
Channel/Axis: 3 Channel/Axis Name: Table Motor
Move Direction Negative
Velocity 3
Ramp Rate 1
Stop Setup Immediate Stop


1
2
8
Home_Table_Motor
9
LatchHomeTable1
PotentiometerPositionMM
Potent_Edge_scale_mm
>
10
LatchHomeTable1
PotentiometerPositionMM
Potent_Edge_scale_mm
<
11
12
PolyPipe.Ax1In3
13
Sensor_fallingEdge
13.1
13.2
14
15


3 4 5 6 7 8 9 10
8 >
9 >
SystemEnable
10 >
11 >
12 >
13 >
13.1 >
13.2 >
14 >
15 >

11

8 > LatchHomeTable1 ( SET )

9 > LatchHomeTable1 ( RST )

10 > Enable

VELOCITY MOVE
Module Name: AMC-1
Channel/Axis: 3 Channel/Axis Name: Table Motor
Move Direction Positive
Velocity 3
Ramp Rate 1
Stop Setup Immediate Stop

11 > ( END )

12 > Sensor_fallingEdge (↑ OUT )

13 > Enable

COPY DATA
Source Destination
PolyPipe.Ax1EncPosition Encoder_Reference
Type Copy Value

13.1 > Enable

MATH
Result Relative_Die_Motor_Position
Numeric Result
Formula
PolyPipe.Ax1EncPosition - Encoder_Reference

13.2 > Enable

MATH
Result cyclic_position
Numeric Result
Formula
Relative_Die_Motor_Position % 10000.0

14 > ( END )

15 > ( END )

| | 1 | 2 |
|---|---|---|
| 16 | | |

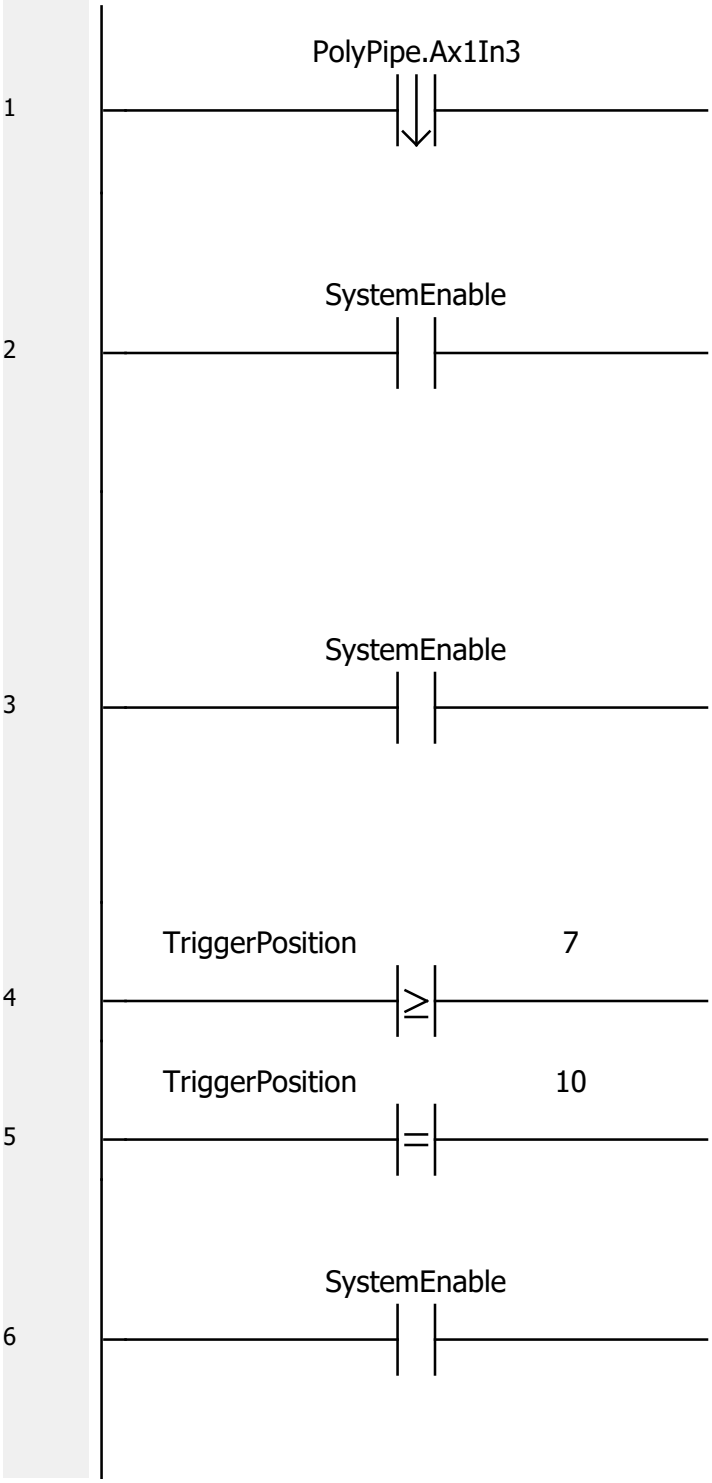
3 4 5 6 7 8 9 10

*16* **>**

11

16 > ——————( END )

1


PolyPipe.Ax1In3
1
SystemEnable
2
SystemEnable
3
TriggerPosition
7
4
TriggerPosition
10
5
SystemEnable
6

2 3

1 >

2 >

3 >

TriggerPosition 9

4 > ≤

5 >

PotentiometerPositionMM Die Index Position1 CV 25

6 > > >

4 5 6 7 8 9 10

1 >

2 >

3 >

4 >

5 >

In_Disable_Range

6 >

1 >

Enable

COPY DATA
Source Destination
0 PolyPipe.Ax1Position
Type Copy Value

2 >

Enable

Top Motor - VELOCITY MOVE
Module Name: AMC-1 In Progress InProg
Channel/Axis: 2 Channel/Axis Name: Top Motor Complete Complet
Move Direction Negative Move Status Statu
Velocity Velocity
Ramp Rate RampRate
Stop Setup Stop at Maximum Decel Rate

3 >

Enable

SIMPLE MOVE
Module Name: AMC-1
Channel/Axis: 1 Channel/Axis Name: Die Motor
Move Addressing Absolute
Direction Negative
Target 1
Velocity PolyPipe.Ax4Velocity
Acceleration DieAccelDecelRate
Deceleration DieAccelDecelRate
Stop Setup Immediate Stop

4 >

In_Disable_Range
( SET )

5 >

In_Disable_Range
( RST )

6 >

Enable

VELOCITY MOVE
Module Name: AMC-1
Channel/Axis: 3 Channel/Axis Name: Table Motor
Move Direction Negative
Velocity Table Motor Velocity
Ramp Rate Table Motor Ramp Rate
Stop Setup Immediate Stop

1 >

gress
:e
2 > s

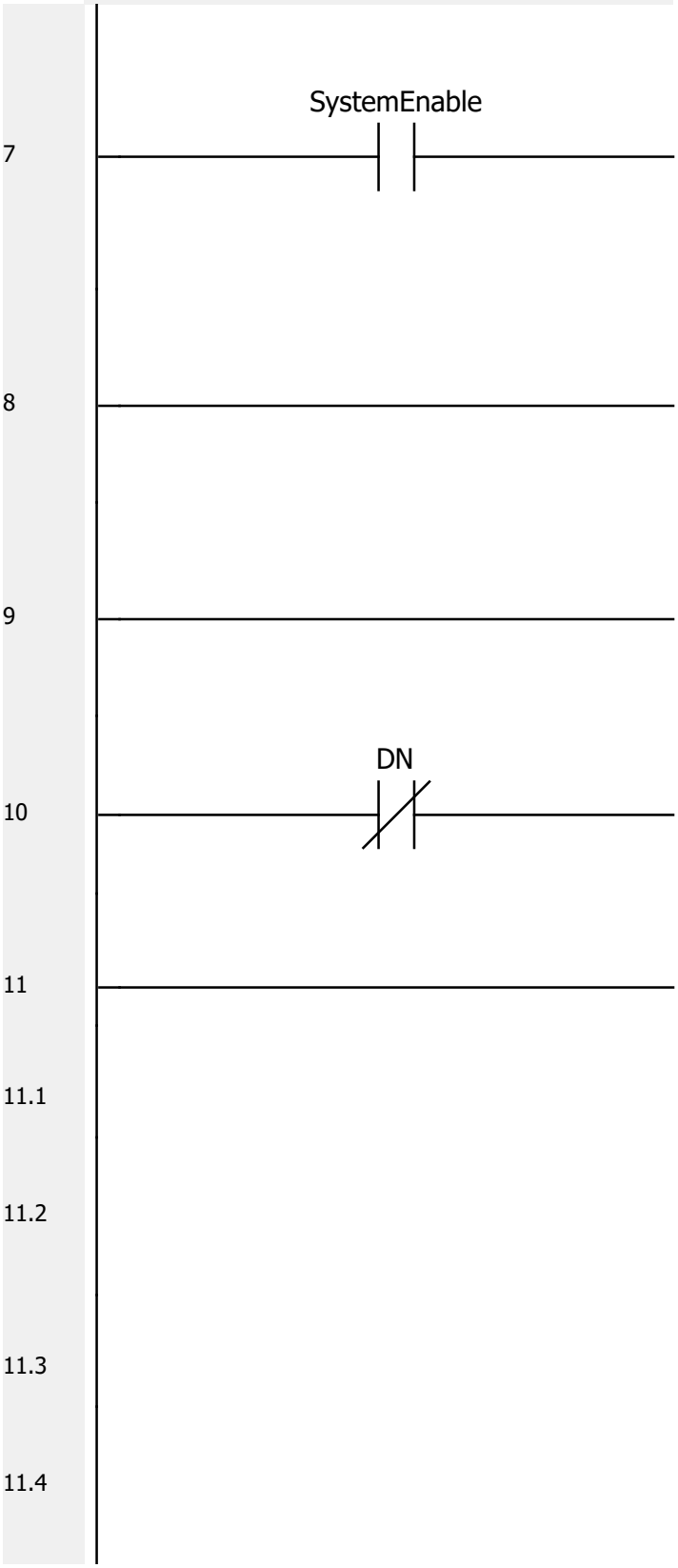
3 >

4 >

5 >

6 >

1



2 3

7 >

PotentiometerPositionMM < Die Index Position1 CV > 25

8 >

9 >

10 >

11 >

PolyPipe.Ax1In3

11.1 >

SystemEnable

11.2 >

CutCycleReset

11.3 >

11.4 >

4 5 6 7 8 9 10
In_Disable_Range
7 >
8 >
9 >
10 >
11 >
11.1 >
SystemEnable
11.2 >
11.3 >
11.4 >

7 > Enable

VELOCITY MOVE
Module Name: AMC-1
Channel/Axis: 3 Channel/Axis Name: Table Motor
Move Direction Positive
Velocity Table Motor Velocity
Ramp Rate Table Motor Ramp Rate
Stop Setup Immediate Stop

8 > Enable

MATH
Result Table Motor Velocity
Numeric Result
Formula
abs( Die Index Position1- PotentiometerPositionMM ) * 5

9 > Enable

MATH
Result Table Motor Ramp Rate
Numeric Result
Formula
Table Motor Velocity * 1.2 * 5

10 > Time Up

SIMPLE TIMER
Preset Value 150 Current Value CV
On Delay Done DN
Time Base .001 sec

11 > StopCutCycle ( SET )

11.1 > TriggerCut ( RST )

11.2 > Enable

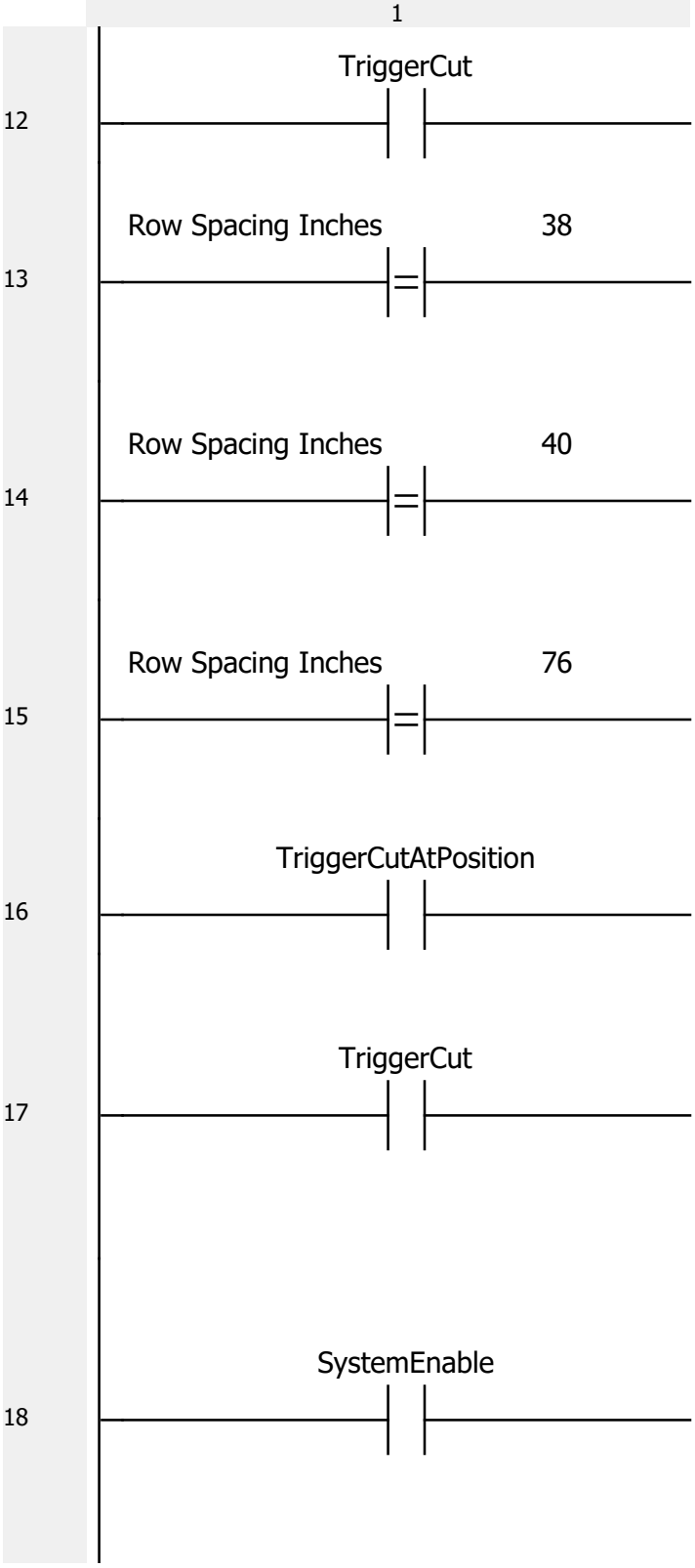
SET POSITION
Module Name: AMC-1
Channel/Axis: 4 Channel/Axis Name: Encoder
Set Position To 0

11.3 > TriggerCutAtPosition ( RST )

11.4 > Enable

SET POSITION
Module Name: AMC-1
Channel/Axis: 1 Channel/Axis Name: Die Motor
Set Position To 0

1


12
TriggerCut
13
Row Spacing Inches
38
14
Row Spacing Inches
40
15
Row Spacing Inches
76
16
TriggerCutAtPosition
17
TriggerCut
18
SystemEnable

2 3

12 >

13 >

14 >

15 >

16 >

17 > StopCutCycle SystemEnable

18 >

4 5 6 7 8 9 10

12 >

13 >

14 >

15 >

16 >

17 >

18 >

StopCutCycle

12 > ( RST )

13 > Enable

MATH
Result TriggerPosition
Numeric Result
Formula
PolyPipe.Ax4EncPosition / 29.29232081

14 > Enable

MATH
Result TriggerPosition
Numeric Result
Formula
PolyPipe.Ax4EncPosition / 29.21018583

15 > Enable

MATH
Result TriggerPosition
Numeric Result
Formula
PolyPipe.Ax4EncPosition / 28.97615132

TriggerCut

16 > ( SET )

17 > Enable

GEAR DRIVETRAIN
Module Name: AMC-1
Drivetrain Segments 1
Encoder as Primary Master Axis Yes
Primary
Master Encoder Slave Die Motor
Ratio 40.3299 Disable

18 > Enable

GEAR DRIVETRAIN
Module Name: AMC-1
Drivetrain Segments 1
Encoder as Primary Master Axis Yes
Primary
Master Encoder Slave Top Motor
Ratio -40.3299 Disable

1

19

20

21

2 3

19 >

20 >

21 >

4 5 6 7 8 9 10

19 >

20 >

21 >

19 > Enable

SCALE NON-LINEAR

Input Edge Sensor Analog Output Edge Position MM

| Input Value | Desired Output |
|---|---|
| 11471 | 0 |
| 13210 | .06 |
| 15694 | 1.48 |
| 21391 | 4.74 |
| 26289 | 7.55 |
| 32208 | 10.94 |
| 39272 | 14.98 |
| 45110 | 18.33 |
| 52250 | 22.42 |
| 59139 | 26.36 |
| 65160 | 29.82 |
| 65565 | 30 |

20 > Enable

SCALE NON-LINEAR

Input R_edge sensor Output R_Edge Position MM

| Input Value | Desired Output |
|---|---|
| 11471 | 0 |
| 13210 | .06 |
| 15694 | 1.48 |
| 21391 | 4.74 |
| 26289 | 7.55 |
| 32208 | 10.94 |
| 39272 | 14.98 |
| 45110 | 18.33 |
| 52250 | 22.42 |
| 59139 | 26.36 |
| 65160 | 29.82 |
| 65565 | 30 |

21 > Enable

SCALE LINEAR

Input PolyPipe.Ax2EncPosition Output Cutting Motor Degrees

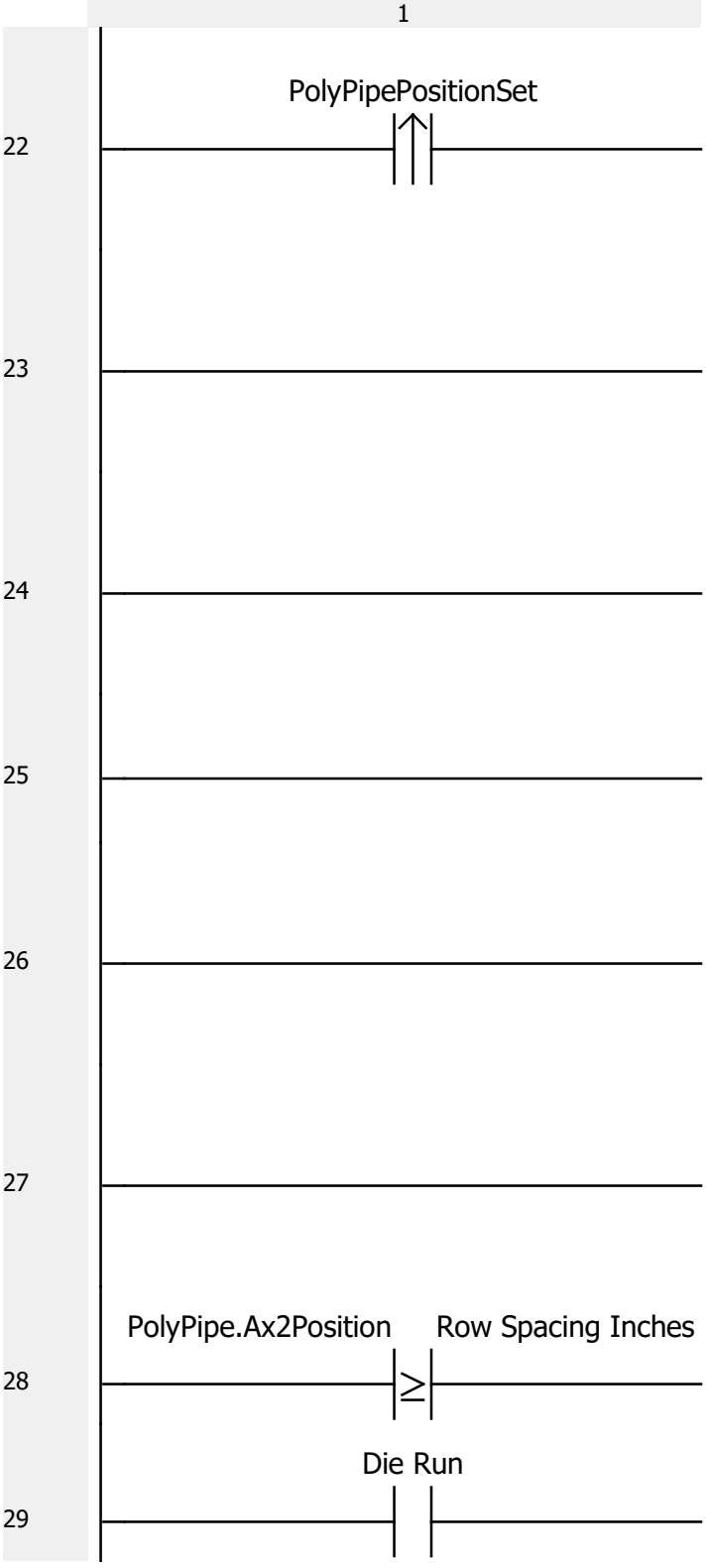
In Min 0

In Max 400

Out Min 360

Out Max 0

1


PolyPipePositionSet
22
23
24
25
26
27
PolyPipe.Ax2Position
Row Spacing Inches
28
Die Run
29

2 3

22 >

23 >

24 >

25 >

26 >

27 >

28 >

29 >

4 5 6 7 8 9 10

22 >

23 >

24 >

25 >

26 >

27 >

28 >

29 >

22 > Enable

COPY DATA
Source: PolyPipe.Ax4EncPosition
Destination: PolyPipePositionCapture
Source: PolyPipe.Ax2EncPosition
Destination: Die Position Capture
Type: Copy Value

23 > Enable

MATH
Result: PolyPipePunchTrigger
Numeric Result
Formula
PolyPipePositionCapture + ((Row Spacing Inches / 6.31974025) * 360

24 > Enable

MATH
Result: Avg_Edge_Position
Numeric Result
Formula
(Edge Position MM + R_Edge Position MM ) / 2

25 > Enable

COMPARE VALUES
Input 1: PolyPipe.Ax4EncPosition
Equal To: PolyPipePositionSet
Input 2: PolyPipePunchTrigger

26 > Enable

MATH
Result: DieAccelDecelRate
Numeric Result
Formula
PolyPipe.Ax4Velocity ^2

27 > Enable

MATH
Result: Die Position Target
Numeric Result
Formula
Die Position Capture + 400

Die Run

28 > ( OUT )

Cutting Motor Off

29 > ( OUT )

22 >

23 >

)

24 >

25 >

26 >

27 >

28 >

29 >

1

30

31 Diameter_MM = 0

32 Diameter_MM ≥ 1.587

33 Diameter_MM ≥ 3.17

34 Diameter_MM ≥ 4.76

35 Diameter_MM ≥ 6.3

36 Diameter_MM ≥ 7.93

37 Diameter_MM ≥ 9.52

2 3

30 >

31 >

32 > Diameter_MM ≤ 1.588

33 > Diameter_MM ≤ 3.18

34 > Diameter_MM ≤ 4.77

35 > Diameter_MM ≤ 6.4

36 > Diameter_MM ≤ 7.94

37 > Diameter_MM ≤ 9.53

4 5 6 7 8 9 10

30 >

31 >

32 >

33 >

34 >

35 >

36 >

37 >

30 > Enable
MATH
Result Diameter_MM
Numeric Result
Formula
(DiameterIn * 25.4)

31 > Enable
COPY DATA
Source Destination
0 Hole Size
Type Copy Value

32 > Enable
COPY DATA
Source Destination
0.306510974 Hole Size
Type Copy Value

33 > Enable
COPY DATA
Source Destination
0.751371601 Hole Size
Type Copy Value

34 > Enable
COPY DATA
Source Destination
1.293767897 Hole Size
Type Copy Value

35 > Enable
COPY DATA
Source Destination
1.918095495 Hole Size
Type Copy Value

36 > Enable
COPY DATA
Source Destination
2.650424332 Hole Size
Type Copy Value

37 > Enable
COPY DATA
Source Destination
3.347503545 Hole Size
Type Copy Value

1

38 Diameter_MM ≥ 11.11

39 Diameter_MM ≥ 12.6

40 Diameter_MM ≥ 14.28

41 Diameter_MM ≥ 15.87

42 Diameter_MM ≥ 17.46

43 Diameter_MM ≥ 19.0

44 Diameter_MM ≥ 20.63

2 3

38 > Diameter_MM ≤ 11.12

39 > Diameter_MM ≤ 12.8

40 > Diameter_MM ≤ 14.29

41 > Diameter_MM ≤ 15.88

42 > Diameter_MM ≤ 17.47

43 > Diameter_MM ≤ 19.1

44 > Diameter_MM ≤ 20.64

4 5 6 7 8 9 10

38 >

39 >

40 >

41 >

42 >

43 >

44 >

11

38 > Enable
COPY DATA
Source Destination
4.180922369 Hole Size
Type Copy Value

39 > Enable
COPY DATA
Source Destination
5.081588517 Hole Size
Type Copy Value

40 > Enable
COPY DATA
Source Destination
5.957857812 Hole Size
Type Copy Value

41 > Enable
COPY DATA
Source Destination
6.959293606 Hole Size
Type Copy Value

42 > Enable
COPY DATA
Source Destination
7.997536308 Hole Size
Type Copy Value

43 > Enable
COPY DATA
Source Destination
9.150634338 Hole Size
Type Copy Value

44 > Enable
COPY DATA
Source Destination
10.31875 Hole Size
Type Copy Value

1

45

46

47

48

RowNumber 0

49

2 3

45 >

46 >

47 >

48 >

49 >

4 5 6 7 8 9 10

45 >

46 >

47 >

48 >

49 >

Adjusted Circle Area Calc75 is where you will find how i made the equation.

45 > Enable

MATH
Result Hole Size
Numeric Result
Formula
(-0.00003 * Diameter_MM^4) + (0.0006 * Diameter_MM ^3) + (0.013 * (Diameter_MM )^2) + 0.1476 * (Diameter_MM ) + 0.4723

46 > Enable

MATH
Result Error Correction
Numeric Result
Formula
0.0312*(Hole Size )^4 - 0.6856*(Hole Size)^3 +5.3482*(Hole Size )^2 - 17.603*(Hole Size ) +20.368

47 > Enable

MATH
Result Corrected Hole Size
Numeric Result
Formula
Hole Size - Error Correction

This function is to find the desired hole size that needs to be cut. Hole Size take into account the area of water flow desired for that row. this is added against the potentiometer which should always be searching for the zero value of the material. There is an offset value added to compensate for the true zero position as it is hard to find true zero by eyeballing it.

48 > Enable

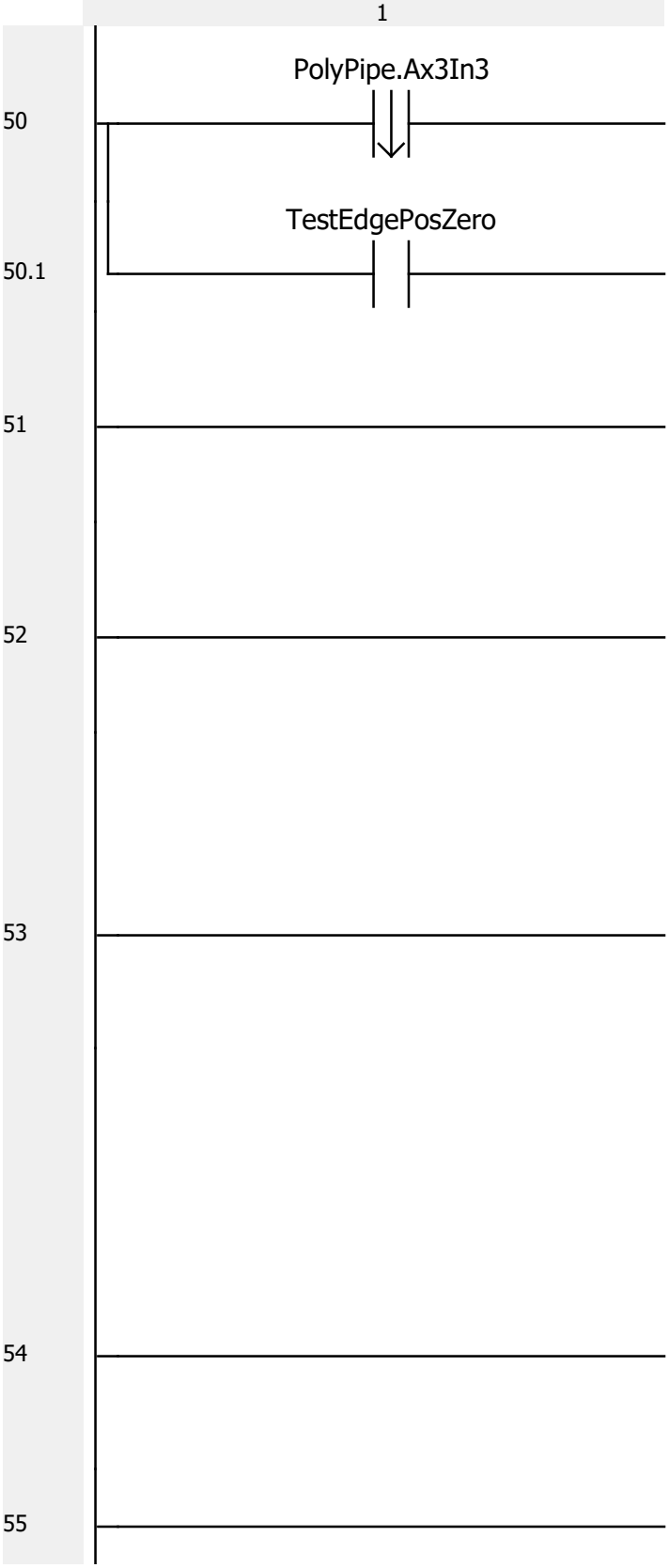
MATH
Result Die Index Target
Numeric Result
Formula
Hole Size + Potent_Edge_scale_mm + 1.809375

This is just for reference. I wanted to see the difference the Error Correction equation is having on the movement of the die.

49 > Enable

MATH
Result Die Index Position1
Numeric Result
Formula
Hole Size + Potent_Edge_scale_mm + 1.809375

1



2 3

50 >

50.1 >

51 >

52 >

53 >

54 >

55 >

4 5 6 7 8 9 10

50 >

50.1 >

51 >

52 >

53 >

54 >

55 >

50 >

Enable

COPY DATA
Source Destination
Avg_Edge_Position Edge Position Reference Zero
Type Copy Value

50.1 >

51 >

Enable

MATH
Result Die Index Velocity
Numeric Result
Formula
abs(PolyPipe.Ax3Position - Die Index Position1)

52 >

Enable

MATH
Result Die Index Ramp Rate
Numeric Result
Formula
Die Index Velocity^2

Reference for die travel. Out min and Out Max are in mm. Gefran potentiometer linear position sensor

53 >

Enable

SCALE LINEAR
Input Die Table Position Analog Output PotentiometerPositionMM
In Min 0
In Max 65535
Out Min 0
Out Max 50.8

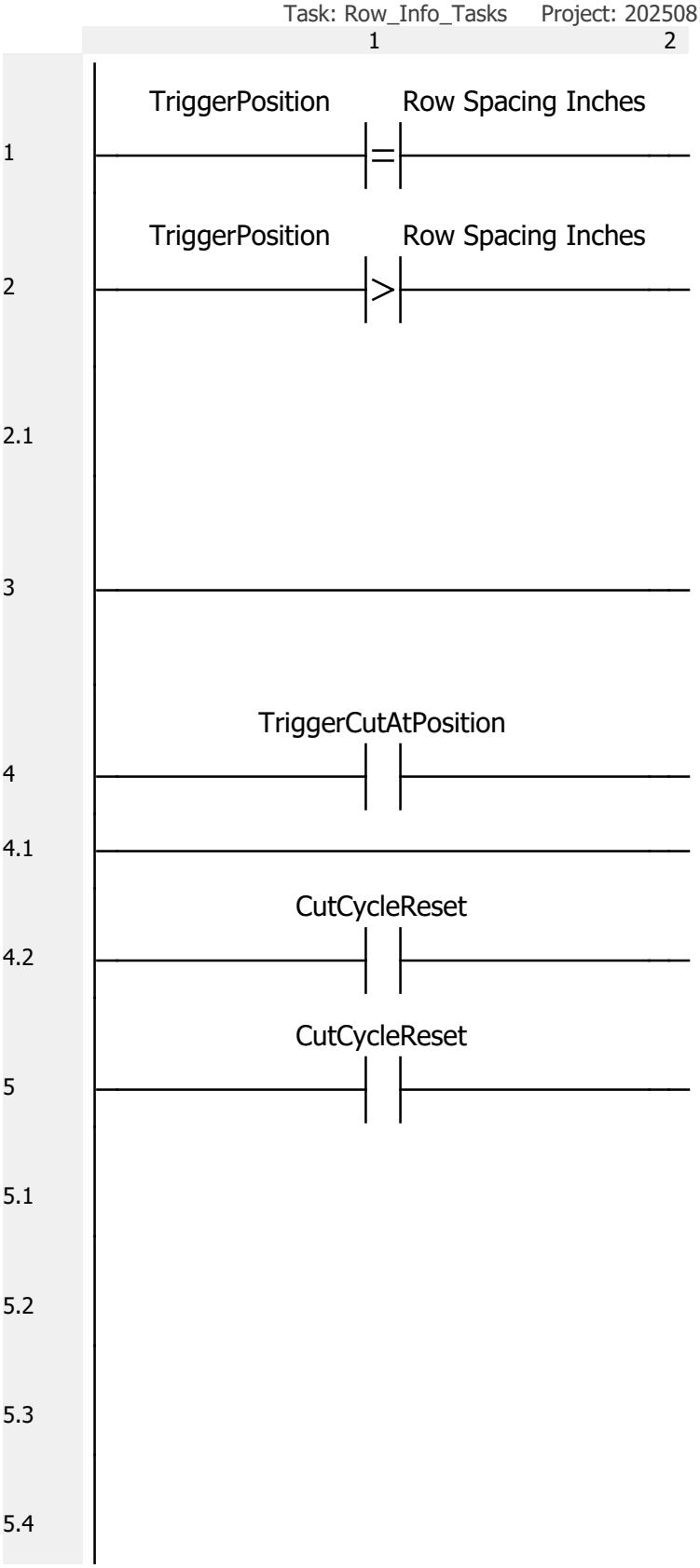
Not sure where the values are coming from yet. However, this is taking the 'Edge Position MM' which is the Edge Sensors analog signal converted to mm. Then it inverts the data so that '+' direction is the same as the '+' direction on the Potentiometer connected to the table.
This allows us to index the die.

54 >

Enable

SCALE LINEAR
Input Avg_Edge_Position Output Potent_Edge_scale_mm
In Min 0.0
In Max 30.0
Out Min 30.92
Out Max 0.92

55 > ( END )


1
2
TriggerPosition
Row Spacing Inches
=
TriggerPosition
Row Spacing Inches
>
TriggerCutAtPosition
CutCycleReset
CutCycleReset
1
2
2.1
3
4
4.1
4.2
5
5.1
5.2
5.3
5.4

3 4 5 6 7

1 >
Row Spacing Inches > 0 SystemEnable

2 >
Row Spacing Inches > 0 SystemEnable

2.1 >

3 >

4 >

4.1 >

4.2 >

5 >

5.1 >

5.2 >

5.3 >

5.4 >

| | 8 | 9 |
|---|---|---|
| 1 > | | |
| 2 > | | |
| 2.1 > | | |
| 3 > | | |
| 4 > | | |
| 4.1 > | | |
| 4.2 > | | |
| 5 > | | |
| 5.1 > | | |
| 5.2 > | | |
| 5.3 > | | |
| 5.4 > | | |

10

1 >

2 >

2.1 >

3 >

4 >

4.1 >

4.2 >

5 >

5.1 >

5.2 >

5.3 >

5.4 >

11

TriggerCutAtPosition

1 > —( SET )

2 > Enable

SET POSITION
Module Name: AMC-1
Channel/Axis: 4 Channel/Axis Name: Encoder
Set Position To 0.1428571429

TriggerCutAtPosition

2.1 > —( RST )

3 > Enable

MATH
Result HolePercentComplete
Numeric Result
Formula
(RowNumber /Total_Row_Number)*100

4 > Cnt Up

4.1 > Cnt Down

4.2 > Reset

COUNTER
Preset Value Total_Row_Number Current Value RowNumber
Don't Reset

Light1

5 > —( RST )

Light2

5.1 > —( RST )

Light3

5.2 > —( RST )

Light4

5.3 > —( RST )

Light5

5.4 > —( RST )

1 2

5.5

5.6

5.7

5.8

5.9

5.10

5.11

5.12

6 RowNumber ≤ 0

6.1

7 RowNumber > 0

7.1

3 4 5 6 7

5.5 >

5.6 >

5.7 >

5.8 >

5.9 >

5.10 >

5.11 >

5.12 >

6 >

6.1 >

RowNumber RowCount

7 > ≤

7.1 >

8 9

5.5 >

5.6 >

5.7 >

5.8 >

5.9 >

5.10 >

5.11 >

5.12 >

6 >

RowNumber 0

6.1 >

7 >

7.1 >

10

5.5 >

5.6 >

5.7 >

5.8 >

5.9 >

5.10 >

5.11 >

5.12 >

6 >

TriggerPosition 9

6.1 > [ > ]

7 >

SystemEnable

7.1 > [ ]

Light6

5.5 > —( RST )

Light7

5.6 > —( RST )

Light8

5.7 > —( RST )

Light9

5.8 > —( RST )

Light10

5.9 > —( RST )

Light11

5.10 > —( RST )

Light12

5.11 > —( RST )

Light13

5.12 > —( RST )

6 > Enable

COPY DATA
Source: Avg_Edge_Position
Destination: Die Index Position1
Type: Copy Value

6.1 > Enable

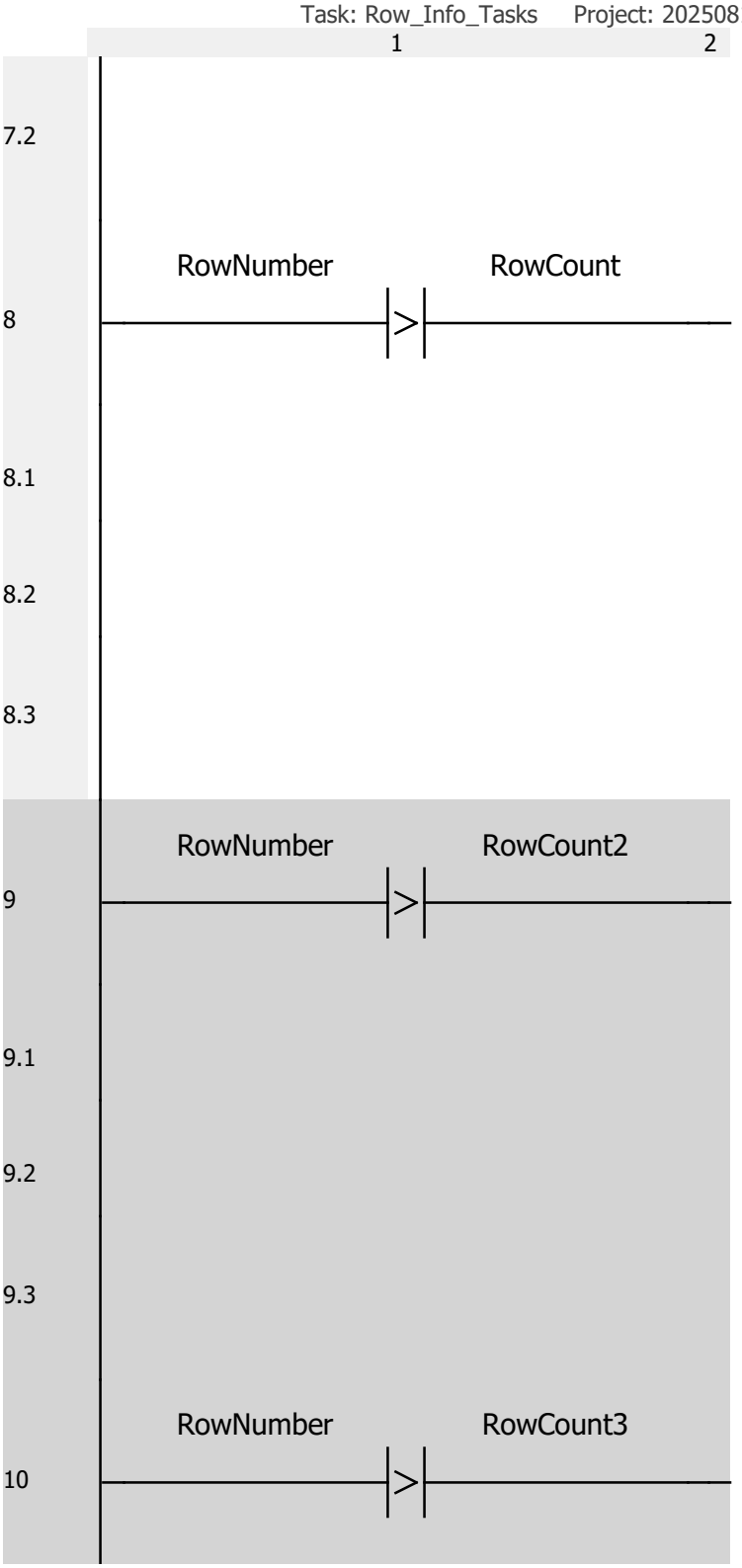
COPY DATA
Source: HoleS1
Destination: DiameterIn
Type: Copy Value

7 > Enable

COPY DATA
Source: HoleS1
Destination: DiameterIn
Type: Copy Value

Light1

7.1 > —( SET )


1
2
7.2
RowNumber
RowCount
8
8.1
8.2
8.3
RowNumber
RowCount2
9
9.1
9.2
9.3
RowNumber
RowCount3
10

3 4 5 6 7

7.2 >

8 >

RowNumber ≤ RowCount2

8.1 >

8.2 >

8.3 >

9 >

RowNumber ≤ RowCount3

9.1 >

9.2 >

9.3 >

10 >

RowNumber ≤ RowCount4

8 9

7.2 >

RowNumber RowCount

=

8 >

8.1 >

8.2 >

RowNumber RowCount2

8.3 >

=

9 >

9.1 >

9.2 >

RowNumber RowCount3

9.3 >

=

10 >

10

TriggerPosition 9

7.2 >

8 >

8.1 >

8.2 >

TriggerPosition 9

8.3 >

9 >

9.1 >

9.2 >

TriggerPosition 9

9.3 >

10 >

11

7.2 >
Enable
COPY DATA
Source Destination
HoleS2 DiameterIn
Type Copy Value

8 >
Enable
COPY DATA
Source Destination
HoleS2 DiameterIn
Type Copy Value

Light2
8.1 > ( SET )

Light1
8.2 > ( RST )

8.3 >
Enable
COPY DATA
Source Destination
HoleS3 DiameterIn
Type Copy Value

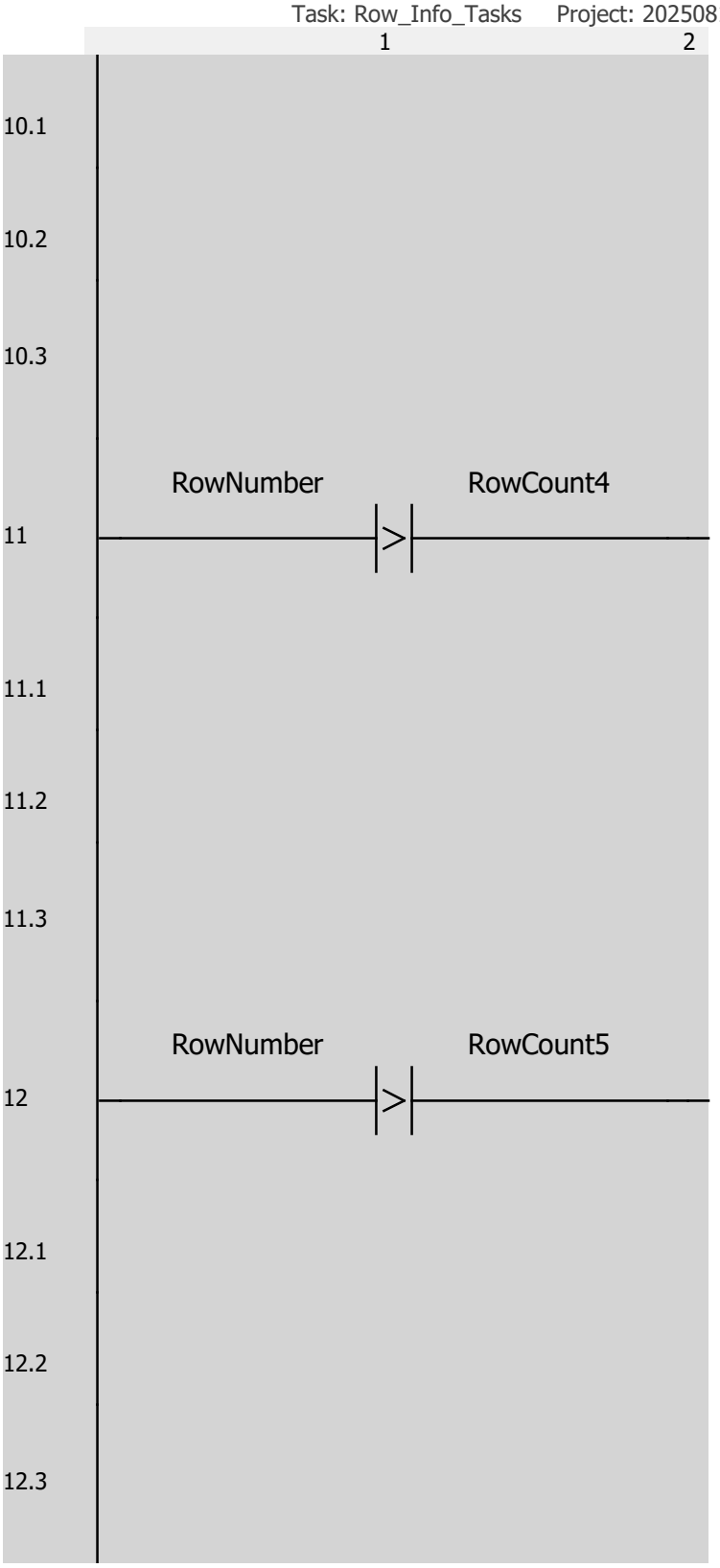
9 >
Enable
COPY DATA
Source Destination
HoleS3 DiameterIn
Type Copy Value

Light3
9.1 > ( SET )

Light2
9.2 > ( RST )

9.3 >
Enable
COPY DATA
Source Destination
HoleS4 DiameterIn
Type Copy Value

10 >
Enable
COPY DATA
Source Destination
HoleS4 DiameterIn
Type Copy Value


1
2
10.1
10.2
10.3
RowNumber
RowCount4
11
11.1
11.2
11.3
RowNumber
RowCount5
12
12.1
12.2
12.3

3 4 5 6 7

10.1 >

10.2 >

10.3 >

RowNumber RowCount5

11 > ≤

11.1 >

11.2 >

11.3 >

RowNumber RowCount6

12 > ≤

12.1 >

12.2 >

12.3 >

8 9

10.1 >

10.2 >

RowNumber RowCount4

10.3 >

11 >

11.1 >

11.2 >

RowNumber RowCount5

11.3 >

12 >

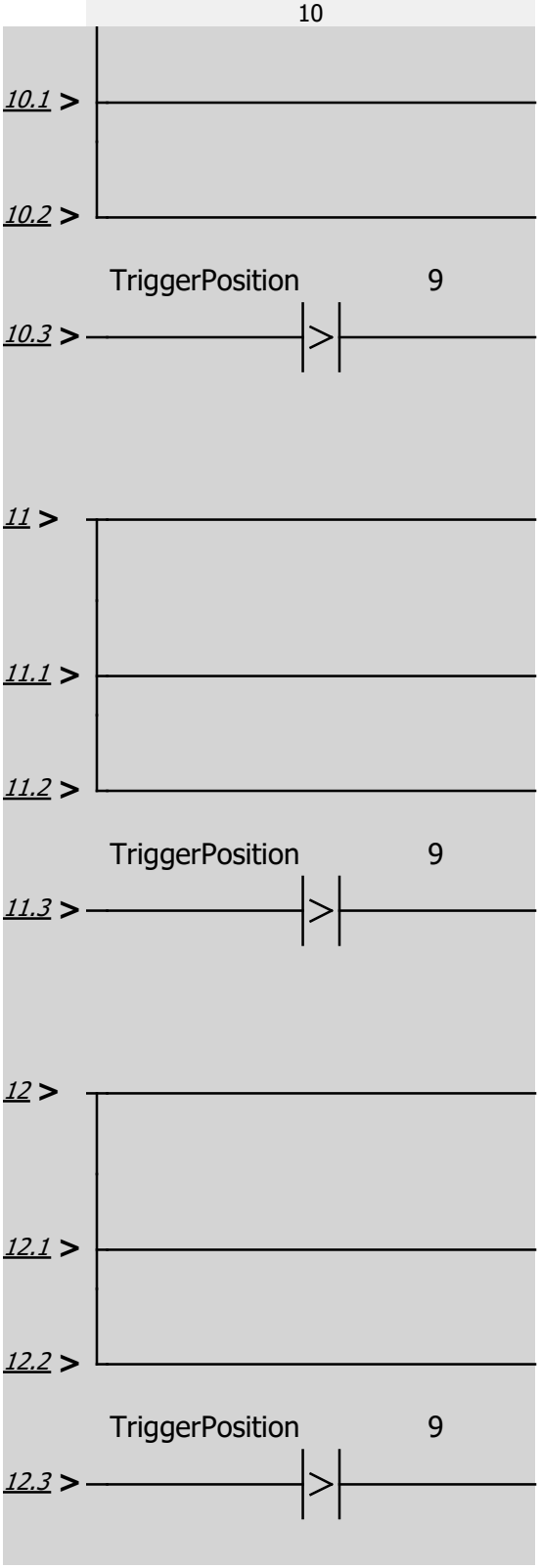
12.1 >

12.2 >

RowNumber RowCount6

12.3 >

10


10.1 >
10.2 >
TriggerPosition
9
10.3 >
11 >
11.1 >
11.2 >
TriggerPosition
9
11.3 >
12 >
12.1 >
12.2 >
TriggerPosition
9
12.3 >

11

Light4

10.1 > ( SET )

Light3

10.2 > ( RST )

10.3 > Enable

COPY DATA
Source Destination
HoleS5 DiameterIn
Type Copy Value

11 > Enable

COPY DATA
Source Destination
HoleS5 DiameterIn
Type Copy Value

Light5

11.1 > ( SET )

Light4

11.2 > ( RST )

11.3 > Enable

COPY DATA
Source Destination
HoleS5 DiameterIn
Type Copy Value

12 > Enable

COPY DATA
Source Destination
HoleS6 DiameterIn
Type Copy Value

Light6

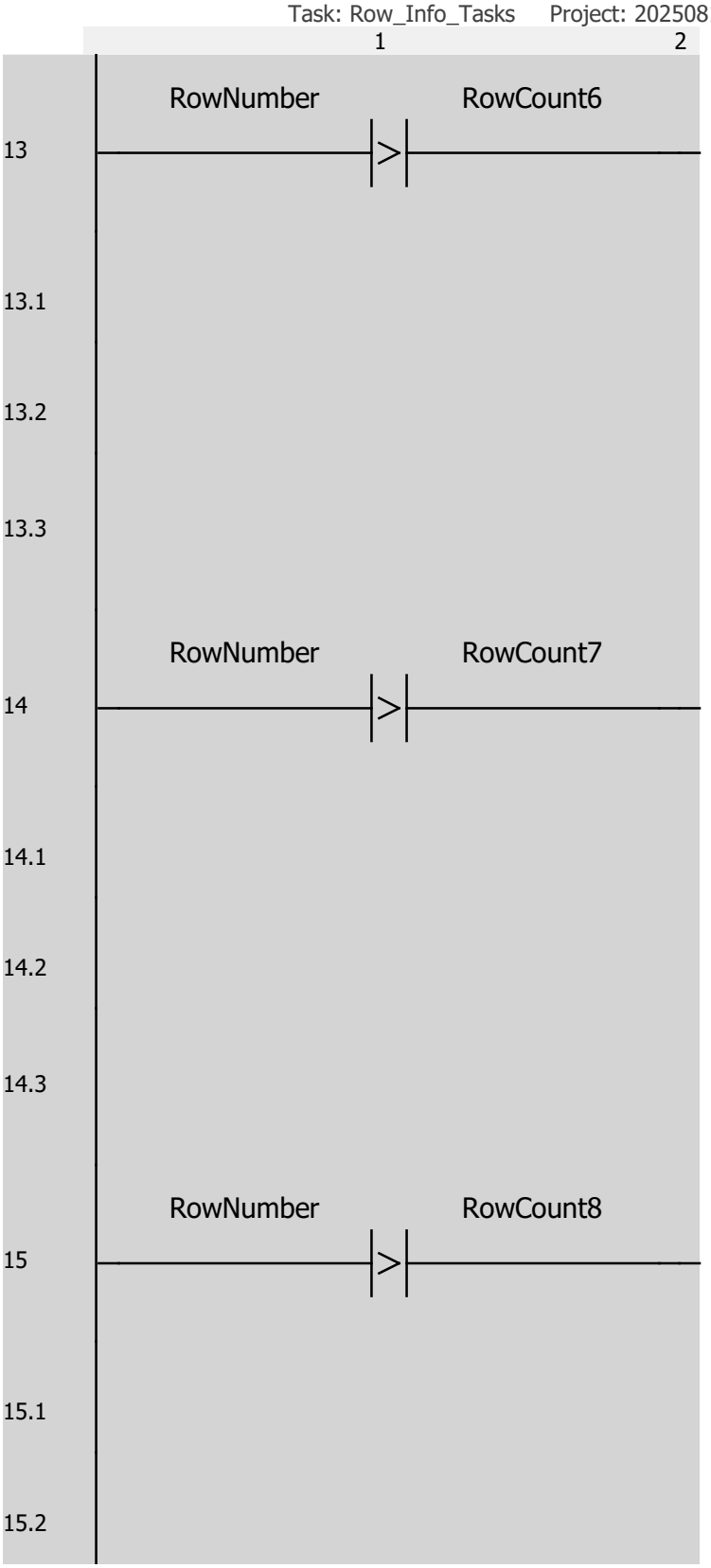
12.1 > ( SET )

Light5

12.2 > ( RST )

12.3 > Enable

COPY DATA
Source Destination
HoleS6 DiameterIn
Type Copy Value


1
2
RowNumber
RowCount6
13
13.1
13.2
13.3
RowNumber
RowCount7
14
14.1
14.2
14.3
RowNumber
RowCount8
15
15.1
15.2

3 4 5 6 7

13 >

RowNumber ≤ RowCount7

13.1 >

13.2 >

13.3 >

14 >

RowNumber ≤ RowCount8

14.1 >

14.2 >

14.3 >

15 >

RowNumber ≤ RowCount9

15.1 >

15.2 >

8 9

13 >

13.1 >

13.2 >

RowNumber RowCount7

13.3 >

14 >

14.1 >

14.2 >

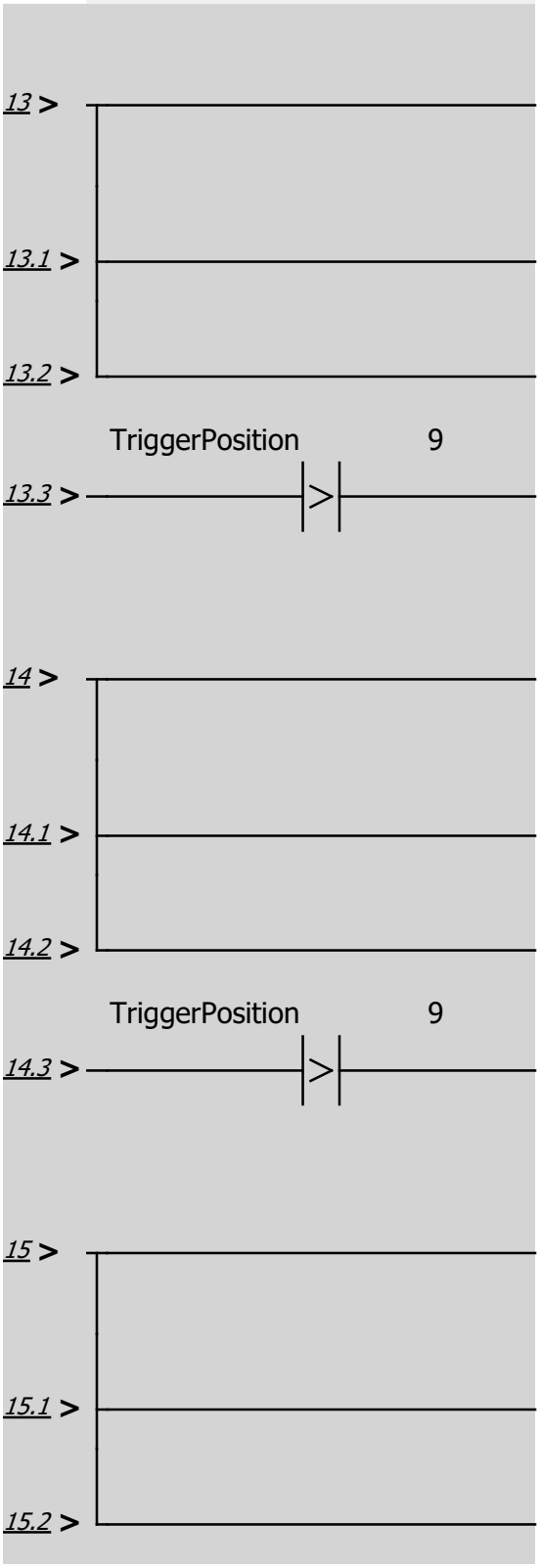
RowNumber RowCount8

14.3 >

15 >

15.1 >

15.2 >


13 >
13.1 >
13.2 >
TriggerPosition 9
13.3 >
14 >
14.1 >
14.2 >
TriggerPosition 9
14.3 >
15 >
15.1 >
15.2 >

11

13 > Enable COPY DATA Source Destination HoleS7 DiameterIn Type Copy Value

13.1 > Light7 ( SET )

13.2 > Light6 ( RST )

13.3 > Enable COPY DATA Source Destination HoleS7 DiameterIn Type Copy Value

14 > Enable COPY DATA Source Destination HoleS8 DiameterIn Type Copy Value

14.1 > Light8 ( SET )

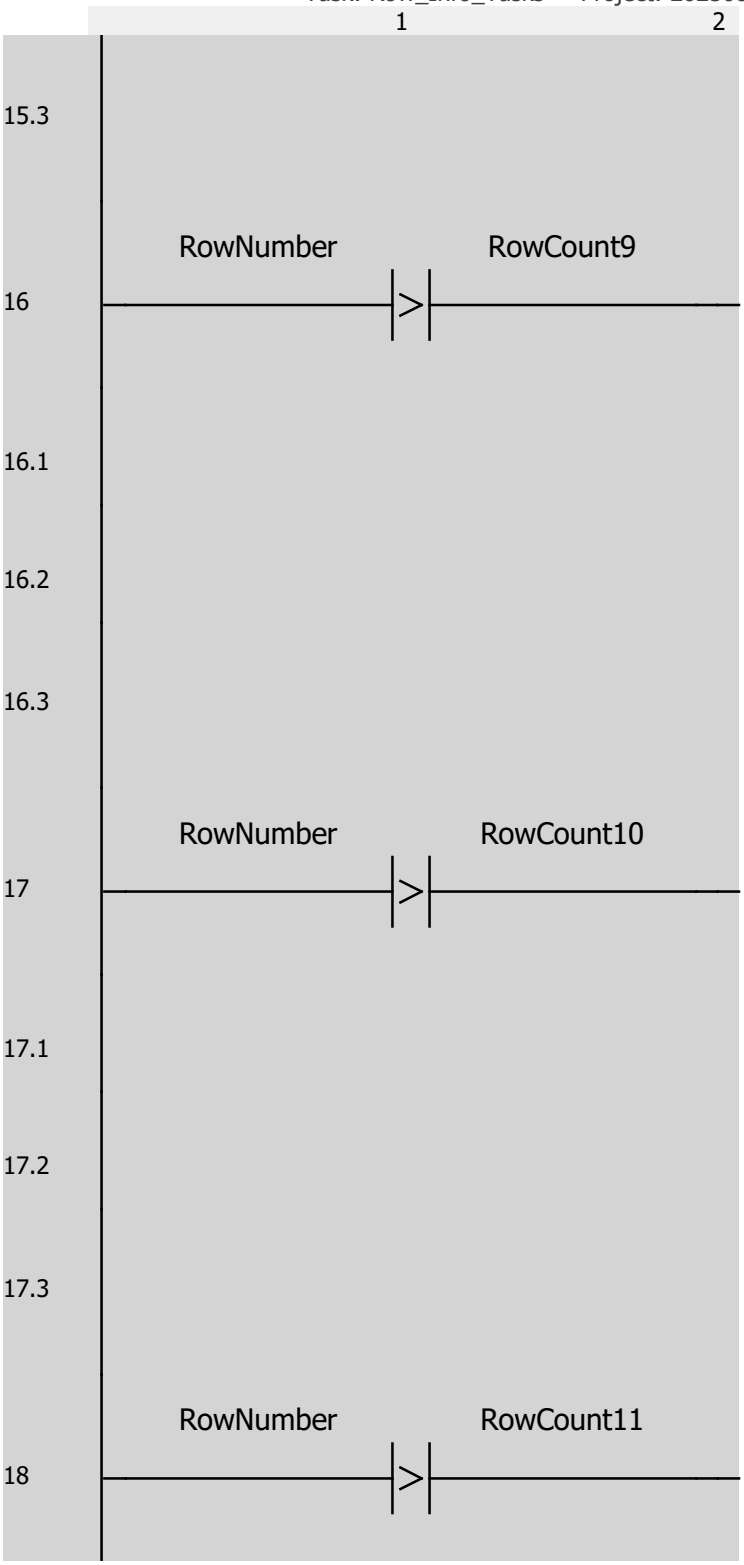
14.2 > Light7 ( RST )

14.3 > Enable COPY DATA Source Destination HoleS8 DiameterIn Type Copy Value

15 > Enable COPY DATA Source Destination HoleS9 DiameterIn Type Copy Value

15.1 > Light9 ( SET )

15.2 > Light8 ( RST )


1
2
15.3
RowNumber
RowCount9
16
16.1
16.2
16.3
RowNumber
RowCount10
17
17.1
17.2
17.3
RowNumber
RowCount11
18

3 4 5 6 7

15.3 >

16 > RowNumber ≤ RowCount10

16.1 >

16.2 >

16.3 >

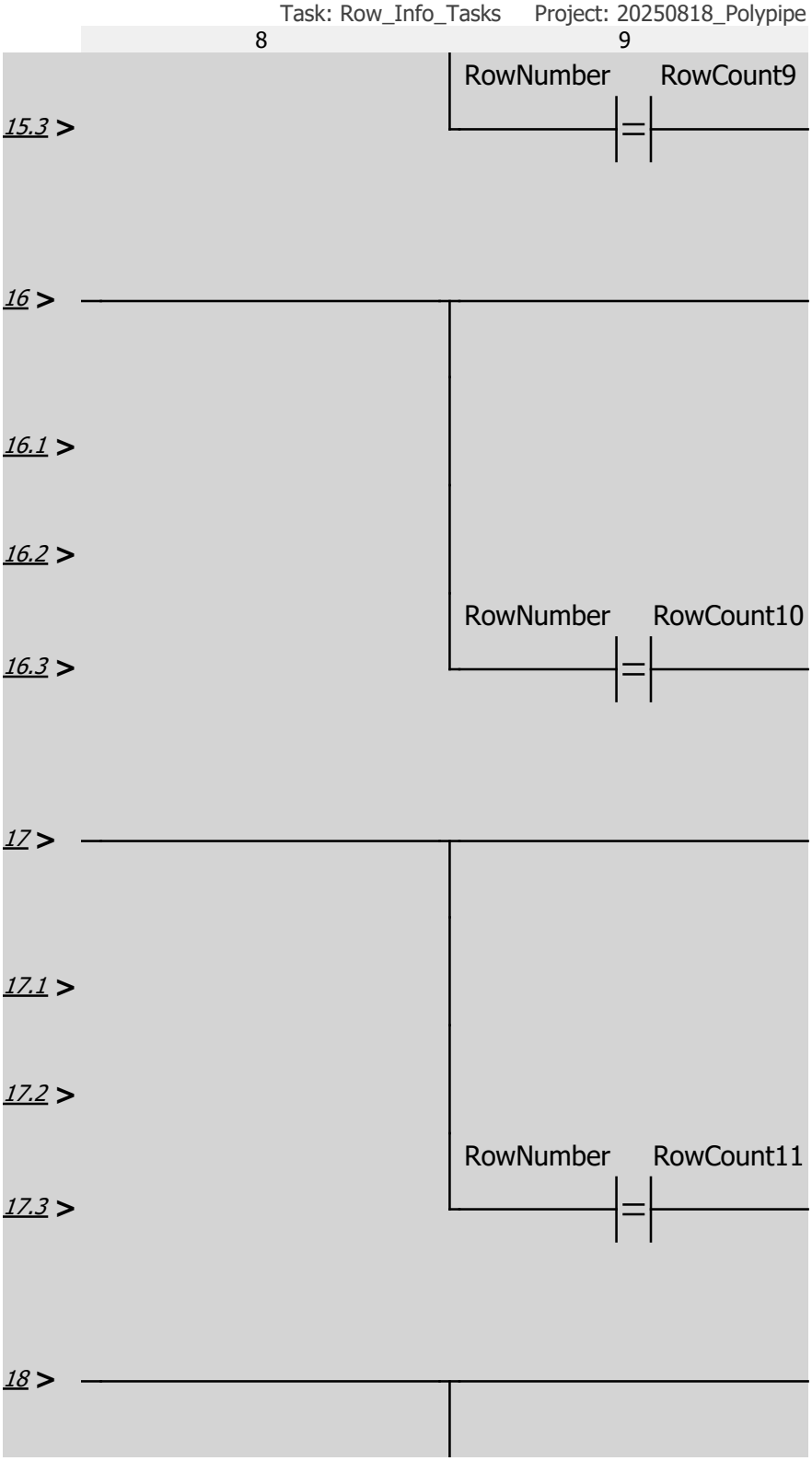
17 > RowNumber ≤ RowCount11

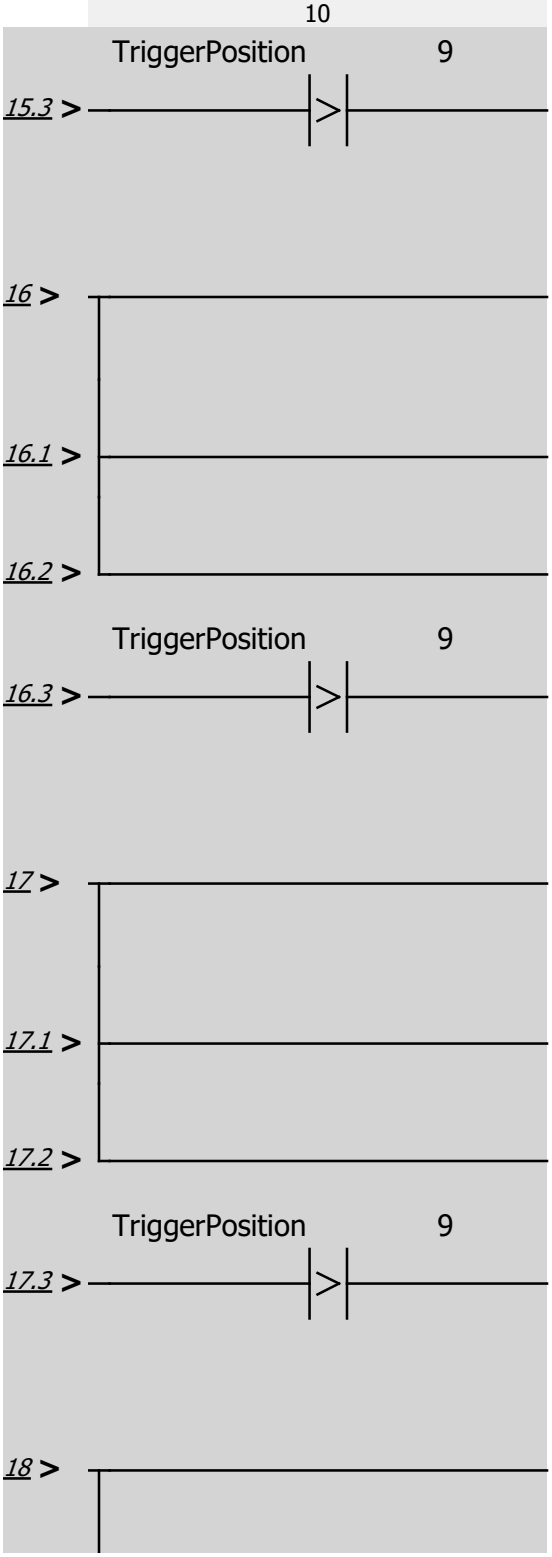
17.1 >

17.2 >

17.3 >

18 > RowNumber ≤ RowCount12


8
9
15.3 >
RowNumber
RowCount9
=
16 >
16.1 >
16.2 >
RowNumber
RowCount10
16.3 >
=
17 >
17.1 >
17.2 >
RowNumber
RowCount11
17.3 >
=
18 >


TriggerPosition 9
15.3 >
16 >
16.1 >
16.2 >
TriggerPosition 9
16.3 >
17 >
17.1 >
17.2 >
TriggerPosition 9
17.3 >
18 >

11

15.3 > Enable
COPY DATA
Source Destination
HoleS9 DiameterIn
Type Copy Value

16 > Enable
COPY DATA
Source Destination
HoleS10 DiameterIn
Type Copy Value

Light10
16.1 > ( SET )

Light9
16.2 > ( RST )

16.3 > Enable
COPY DATA
Source Destination
HoleS10 DiameterIn
Type Copy Value

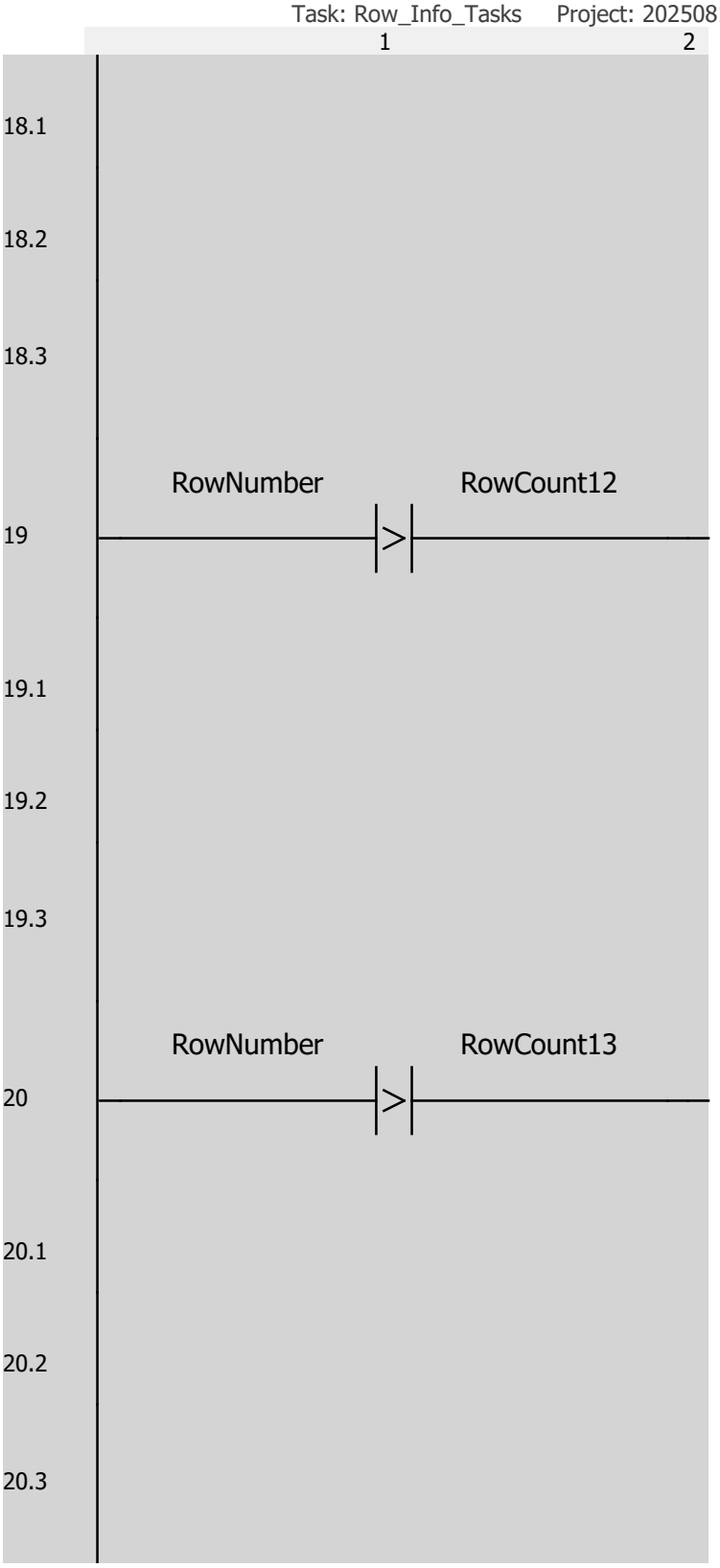
17 > Enable
COPY DATA
Source Destination
HoleS11 DiameterIn
Type Copy Value

Light11
17.1 > ( SET )

Light10
17.2 > ( RST )

17.3 > Enable
COPY DATA
Source Destination
HoleS11 DiameterIn
Type Copy Value

18 > Enable
COPY DATA
Source Destination
HoleS12 DiameterIn
Type Copy Value


1
2
18.1
18.2
18.3
RowNumber
RowCount12
19
19.1
19.2
19.3
RowNumber
RowCount13
20
20.1
20.2
20.3

3 4 5 6 7

18.1 >

18.2 >

18.3 >

19 > RowNumber ≤ RowCount13

19.1 >

19.2 >

19.3 >

20 > RowNumber ≤ RowCount14

20.1 >

20.2 >

20.3 >

8 9

18.1 >

18.2 >

RowNumber RowCount12

18.3 >

19 >

19.1 >

19.2 >

RowNumber RowCount13

19.3 >

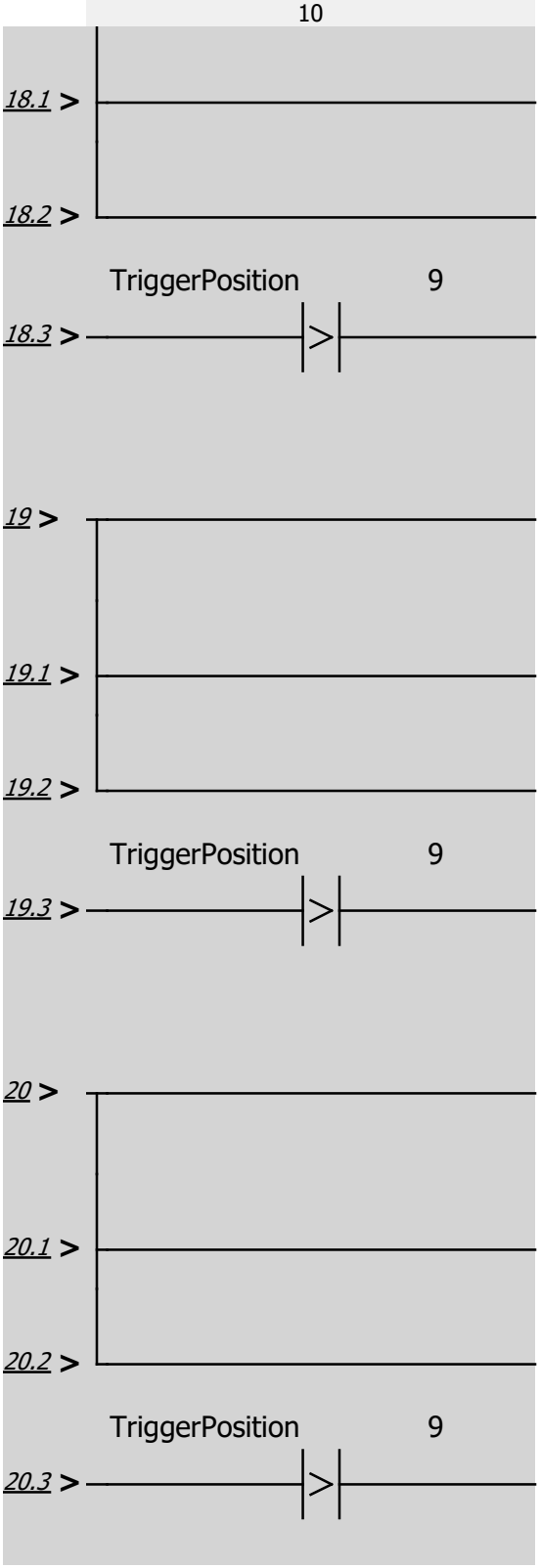
20 >

20.1 >

20.2 >

RowNumber RowCount14

20.3 >


18.1 >
18.2 >
TriggerPosition 9
18.3 >
19 >
19.1 >
19.2 >
TriggerPosition 9
19.3 >
20 >
20.1 >
20.2 >
TriggerPosition 9
20.3 >

11

Light12

18.1 > —( SET )

Light11

18.2 > —( RST )

18.3 > — Enable

COPY DATA
Source Destination
HoleS12 DiameterIn
Type Copy Value

19 > — Enable

COPY DATA
Source Destination
HoleS13 DiameterIn
Type Copy Value

Light13

19.1 > —( SET )

Light12

19.2 > —( RST )

19.3 > — Enable

COPY DATA
Source Destination
HoleS13 DiameterIn
Type Copy Value

20 > — Enable

COPY DATA
Source Destination
HoleS14 DiameterIn
Type Copy Value

Light14

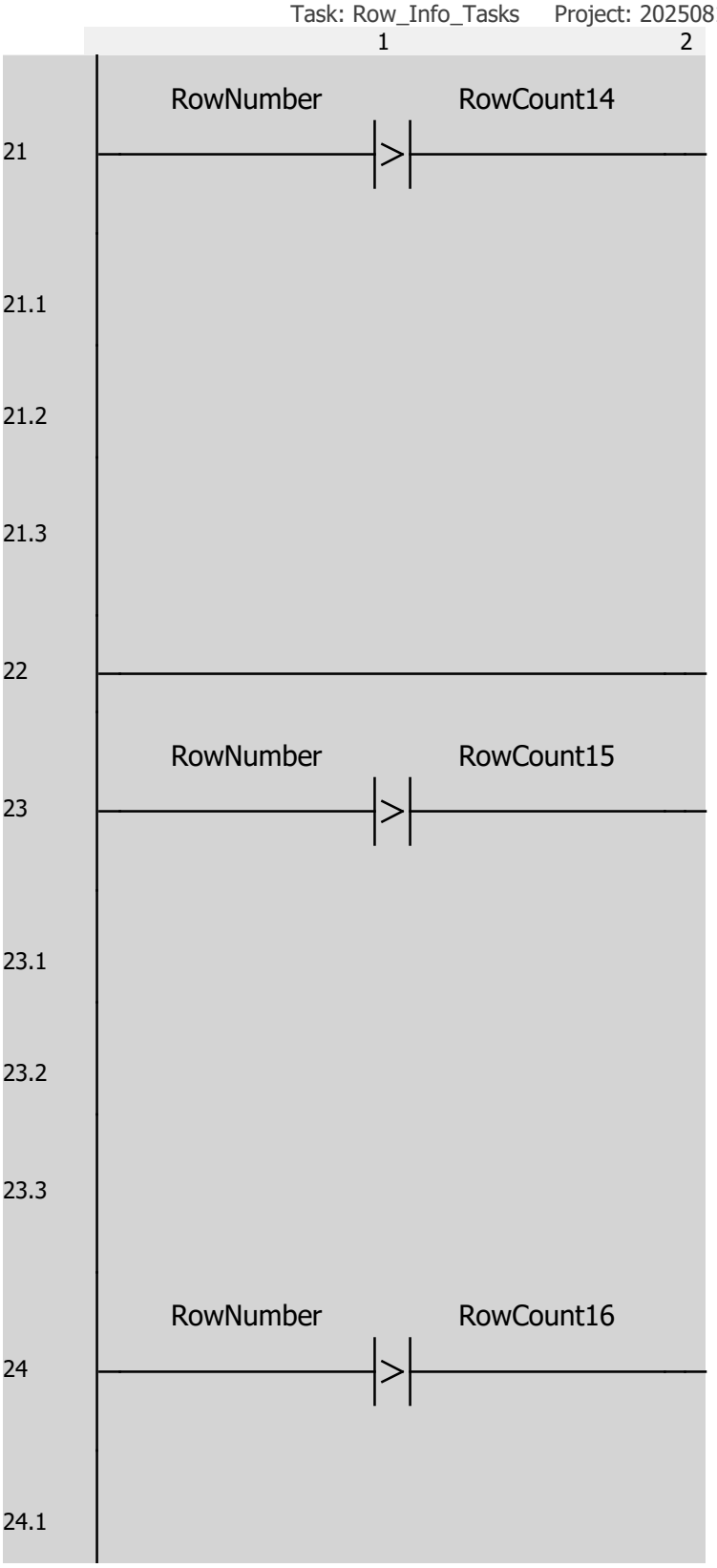
20.1 > —( SET )

Light13

20.2 > —( RST )

20.3 > — Enable

COPY DATA
Source Destination
HoleS14 DiameterIn
Type Copy Value


1
2
RowNumber
RowCount14
21
21.1
21.2
21.3
22
RowNumber
RowCount15
23
23.1
23.2
23.3
RowNumber
RowCount16
24
24.1

3 4 5 6 7

21 >

RowNumber ≤ RowCount15

21.1 >

21.2 >

21.3 >

22 >

23 >

RowNumber ≤ RowCount16

23.1 >

23.2 >

23.3 >

24 >

RowNumber ≤ RowCount17

24.1 >

8 9

21 >

21.1 >

21.2 >

RowNumber RowCount15

21.3 >

22 >

23 >

23.1 >

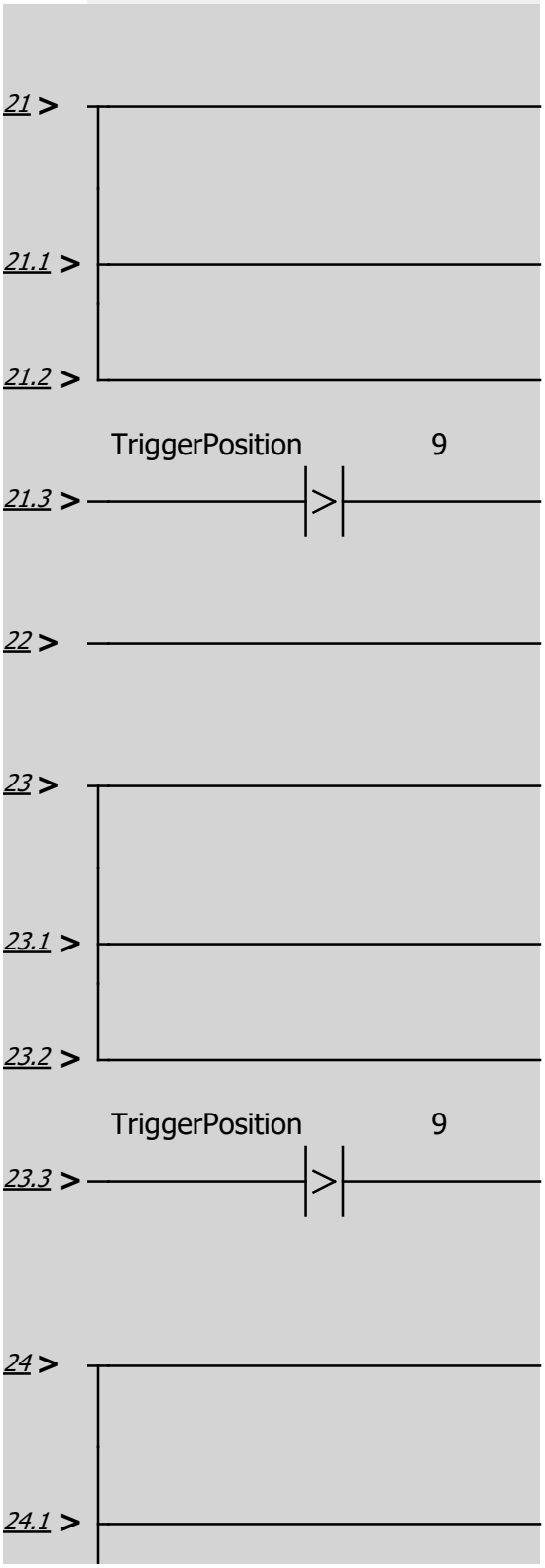
23.2 >

RowNumber RowCount16

23.3 >

24 >

24.1 >


21 >
21.1 >
21.2 >
TriggerPosition 9
21.3 >
22 >
23 >
23.1 >
23.2 >
TriggerPosition 9
23.3 >
24 >
24.1 >

11

21 > Enable COPY DATA
Source Destination
HoleS15 DiameterIn
Type Copy Value

Light15
21.1 > ( SET )

Light14
21.2 > ( RST )

21.3 > Enable COPY DATA
Source Destination
HoleS15 DiameterIn
Type Copy Value

22 > ( NOP )

23 > Enable COPY DATA
Source Destination
HoleS16 DiameterIn
Type Copy Value

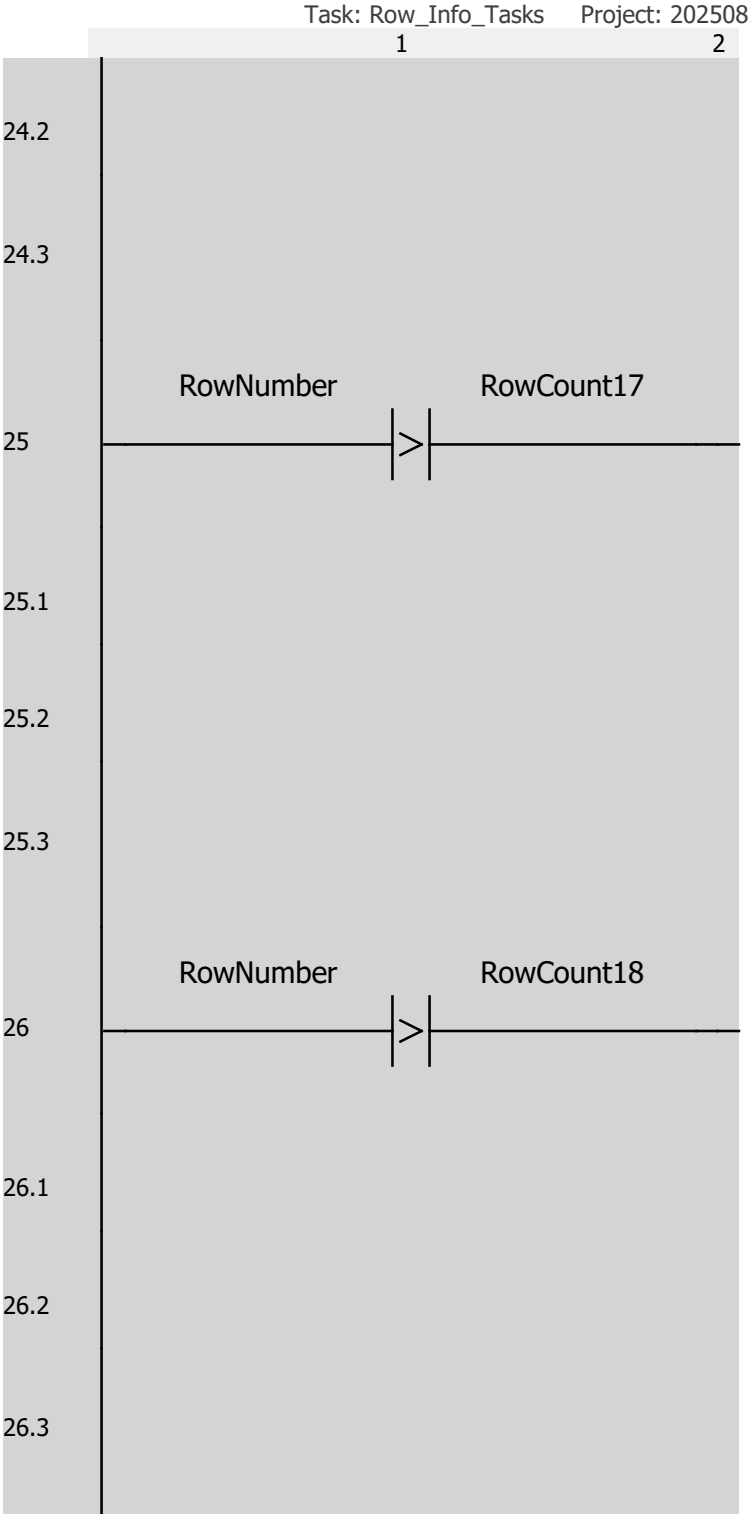
Light16
23.1 > ( SET )

Light15
23.2 > ( RST )

23.3 > Enable COPY DATA
Source Destination
HoleS16 DiameterIn
Type Copy Value

24 > Enable COPY DATA
Source Destination
HoleS17 DiameterIn
Type Copy Value

Light17
24.1 > ( SET )


1
2
24.2
24.3
RowNumber
RowCount17
25
25.1
25.2
25.3
RowNumber
RowCount18
26
26.1
26.2
26.3

3 4 5 6 7

24.2 >

24.3 >

25 >

RowNumber ≤ RowCount18

25.1 >

25.2 >

25.3 >

26 >

RowNumber ≤ RowCount19

26.1 >

26.2 >

26.3 >

8 9

24.2 >

RowNumber RowCount17

24.3 >

25 >

25.1 >

25.2 >

RowNumber RowCount18

25.3 >

26 >

26.1 >

26.2 >

RowNumber RowCount19

26.3 >

24.2 >

TriggerPosition 9

24.3 > >

25 >

25.1 >

25.2 >

TriggerPosition 9

25.3 > >

26 >

26.1 >

26.2 >

TriggerPosition 9

26.3 > >

11

Light16

24.2 > ( RST )

24.3 > Enable

COPY DATA
Source Destination
HoleS17 DiameterIn
Type Copy Value

25 > Enable

COPY DATA
Source Destination
HoleS18 DiameterIn
Type Copy Value

Light18

25.1 > ( SET )

Light17

25.2 > ( RST )

25.3 > Enable

COPY DATA
Source Destination
HoleS18 DiameterIn
Type Copy Value

26 > Enable

COPY DATA
Source Destination
HoleS19 DiameterIn
Type Copy Value

Light19

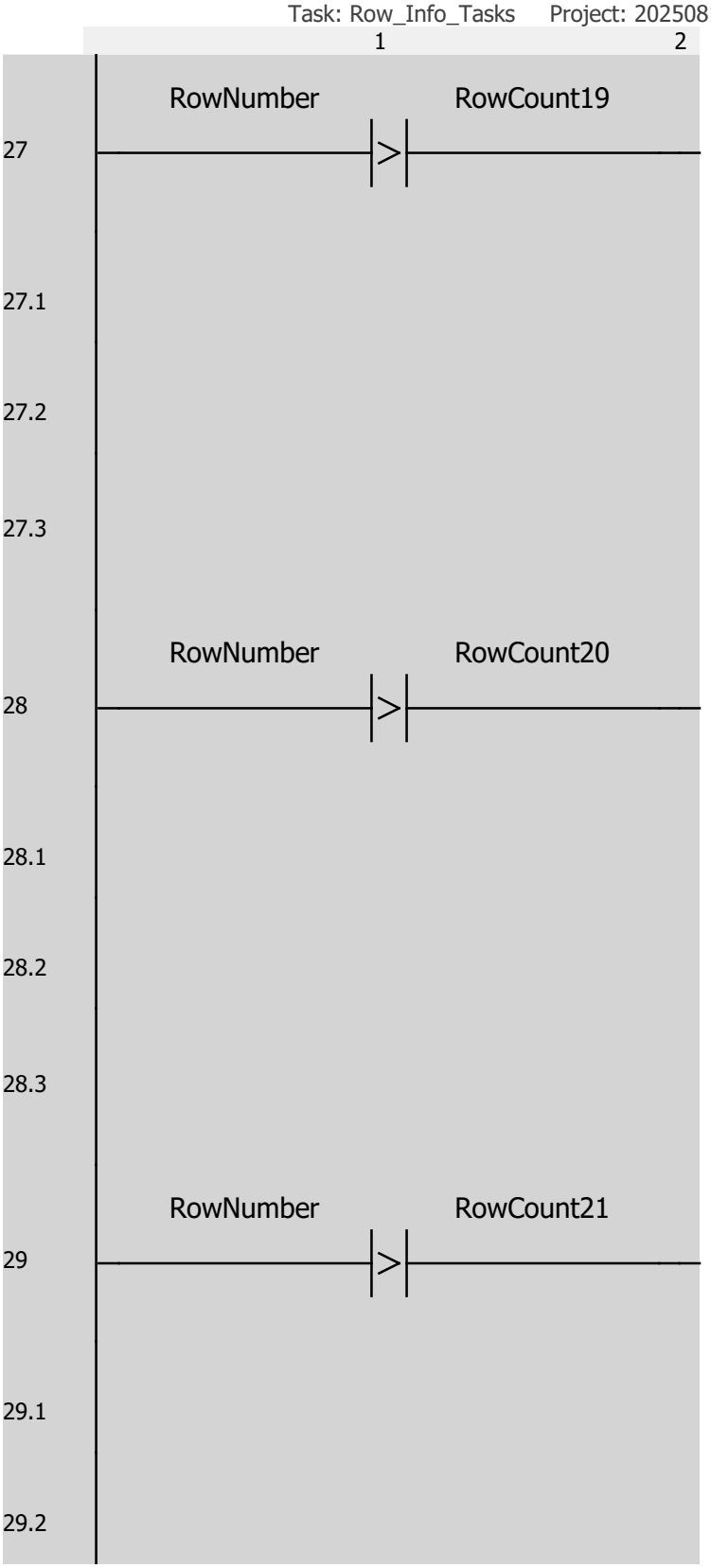
26.1 > ( SET )

Light18

26.2 > ( RST )

26.3 > Enable

COPY DATA
Source Destination
HoleS19 DiameterIn
Type Copy Value


1
2
RowNumber
RowCount19
27
27.1
27.2
27.3
RowNumber
RowCount20
28
28.1
28.2
28.3
RowNumber
RowCount21
29
29.1
29.2

3 4 5 6 7

27 >

RowNumber RowCount20

27.1 >

27.2 >

27.3 >

28 >

RowNumber RowCount21

28.1 >

28.2 >

28.3 >

29 >

RowNumber RowCount22

29.1 >

29.2 >

8 9

27 >

27.1 >

27.2 >

RowNumber RowCount20

27.3 >

28 >

28.1 >

28.2 >

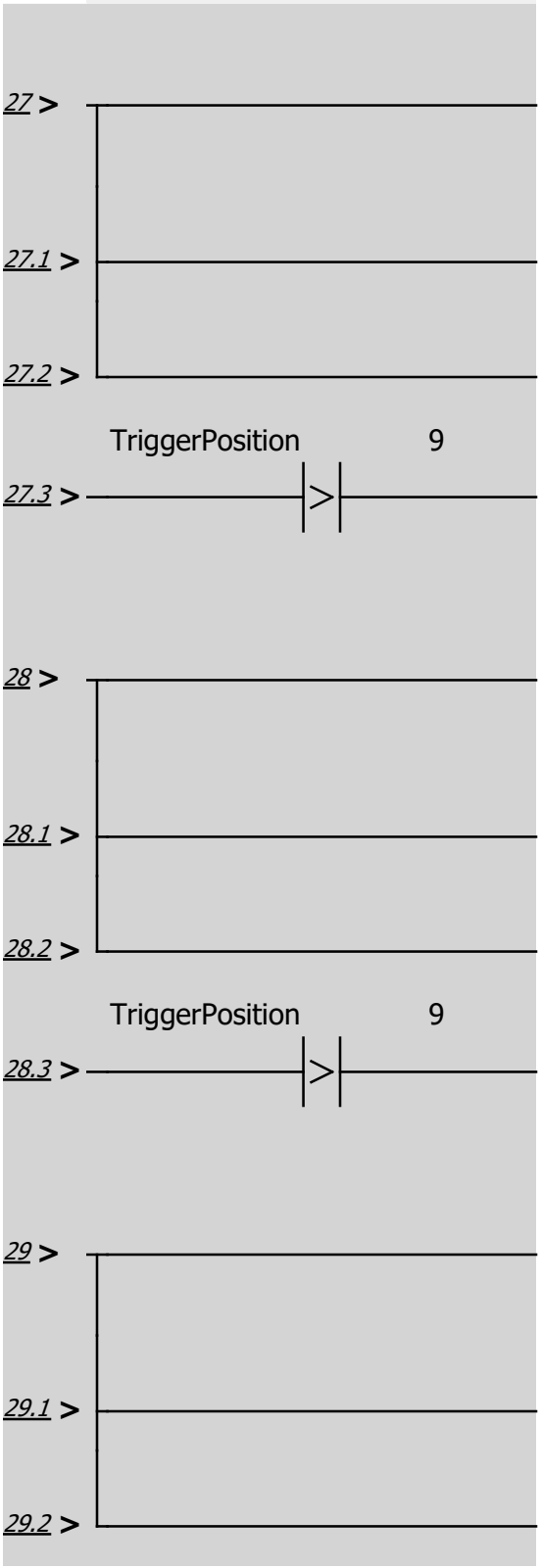
RowNumber RowCount21

28.3 >

29 >

29.1 >

29.2 >


27 >
27.1 >
27.2 >
TriggerPosition 9
27.3 >
28 >
28.1 >
28.2 >
TriggerPosition 9
28.3 >
29 >
29.1 >
29.2 >

11

27 > Enable COPY DATA
Source Destination
HoleS20 DiameterIn
Type Copy Value

27.1 > Light20 ( SET )

27.2 > Light19 ( RST )

27.3 > Enable COPY DATA
Source Destination
HoleS20 DiameterIn
Type Copy Value

28 > Enable COPY DATA
Source Destination
HoleS21 DiameterIn
Type Copy Value

28.1 > Light21 ( SET )

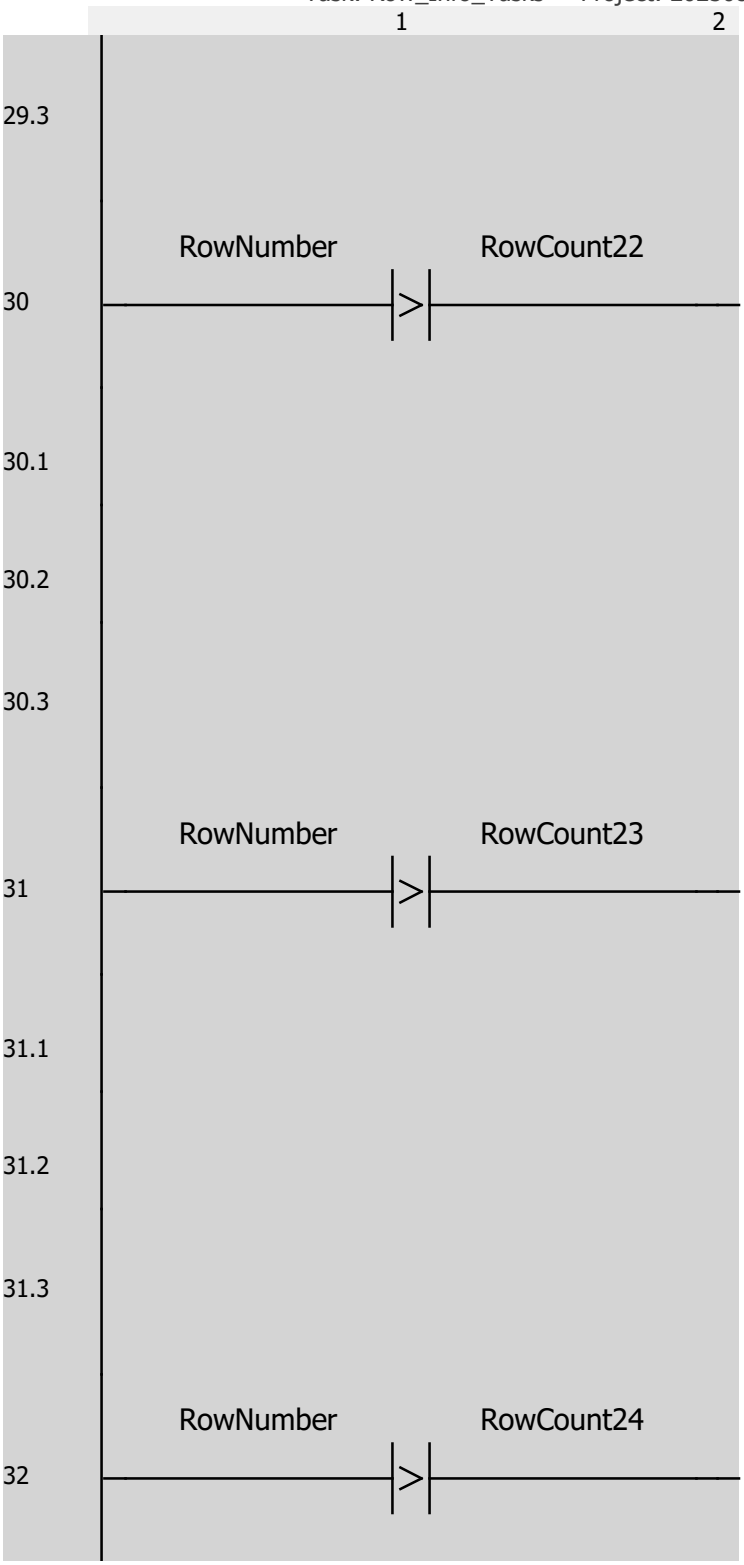
28.2 > Light20 ( RST )

28.3 > Enable COPY DATA
Source Destination
HoleS21 DiameterIn
Type Copy Value

29 > Enable COPY DATA
Source Destination
HoleS22 DiameterIn
Type Copy Value

29.1 > Light21 ( SET )

29.2 > Light20 ( RST )


1
2
29.3
RowNumber
RowCount22
30
30.1
30.2
30.3
RowNumber
RowCount23
31
31.1
31.2
31.3
RowNumber
RowCount24
32

3 4 5 6 7

29.3 >

30 > RowNumber ≤ RowCount23

30.1 >

30.2 >

30.3 >

31 > RowNumber ≤ RowCount24

31.1 >

31.2 >

31.3 >

32 > RowNumber ≤ RowCount25

8 9

29.3 >

RowNumber RowCount21

=

30 >

30.1 >

30.2 >

RowNumber RowCount21

30.3 >

=

31 >

31.1 >

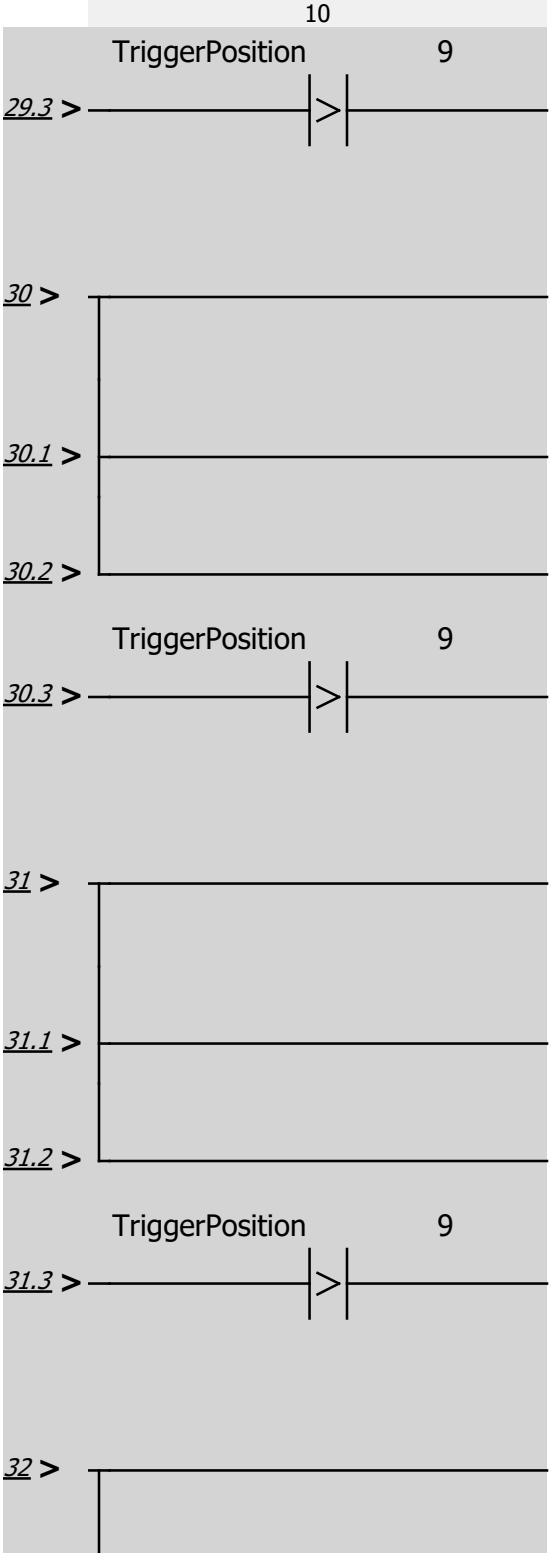
31.2 >

RowNumber RowCount21

31.3 >

=

32 >


TriggerPosition 9
29.3 >
30 >
30.1 >
30.2 >
TriggerPosition 9
30.3 >
31 >
31.1 >
31.2 >
TriggerPosition 9
31.3 >
32 >

11

29.3 > Enable
COPY DATA
Source Destination
HoleS22 DiameterIn
Type Copy Value

30 > Enable
COPY DATA
Source Destination
HoleS23 DiameterIn
Type Copy Value

Light21
30.1 > ( SET )

Light20
30.2 > ( RST )

30.3 > Enable
COPY DATA
Source Destination
HoleS23 DiameterIn
Type Copy Value

31 > Enable
COPY DATA
Source Destination
HoleS24 DiameterIn
Type Copy Value

Light21
31.1 > ( SET )

Light20
31.2 > ( RST )

31.3 > Enable
COPY DATA
Source Destination
HoleS24 DiameterIn
Type Copy Value

32 > Enable
COPY DATA
Source Destination
HoleS25 DiameterIn
Type Copy Value

1 2

32.1

32.2

32.3

33 RowNumber > RowCount25

33.1

33.2

33.3

34 RowNumber > RowCount26

34.1

34.2

34.3

3 4 5 6 7

32.1 >

32.2 >

32.3 >

33 > RowNumber ≤ RowCount26

33.1 >

33.2 >

33.3 >

34 > RowNumber ≤ RowCount27

34.1 >

34.2 >

34.3 >

8 9

32.1 >

32.2 >

RowNumber RowCount21

32.3 >

33 >

33.1 >

33.2 >

RowNumber RowCount21

33.3 >

34 >

34.1 >

34.2 >

RowNumber RowCount21

34.3 >

32.1 >

32.2 >

TriggerPosition 9

32.3 >

33 >

33.1 >

33.2 >

TriggerPosition 9

33.3 >

34 >

34.1 >

34.2 >

TriggerPosition 9

34.3 >

11

Light21

32.1 > —( SET )

Light20

32.2 > —( RST )

32.3 > Enable COPY DATA
Source Destination
HoleS25 DiameterIn
Type Copy Value

33 > Enable COPY DATA
Source Destination
HoleS26 DiameterIn
Type Copy Value

Light21

33.1 > —( SET )

Light20

33.2 > —( RST )

33.3 > Enable COPY DATA
Source Destination
HoleS26 DiameterIn
Type Copy Value

34 > Enable COPY DATA
Source Destination
HoleS27 DiameterIn
Type Copy Value

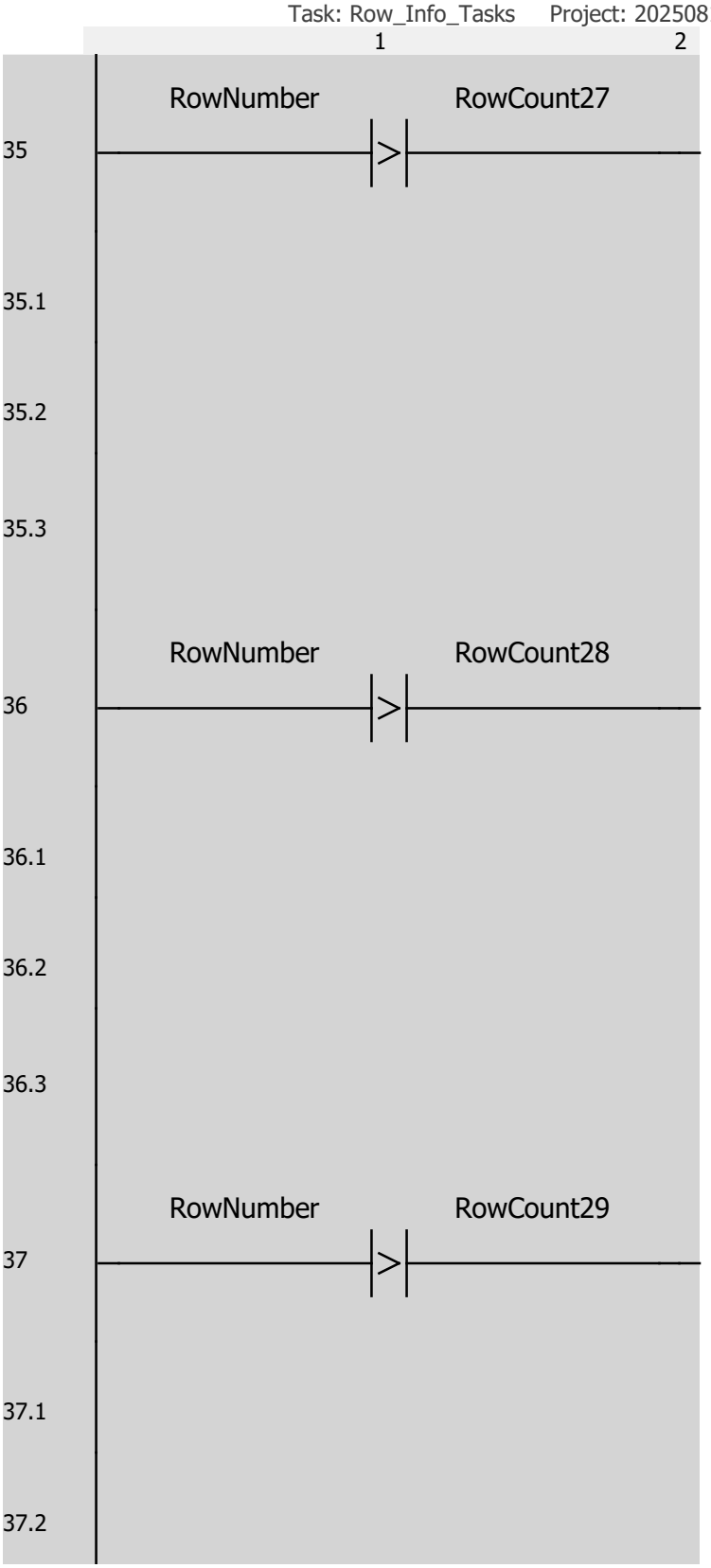
Light21

34.1 > —( SET )

Light20

34.2 > —( RST )

34.3 > Enable COPY DATA
Source Destination
HoleS27 DiameterIn
Type Copy Value


1
2
RowNumber
RowCount27
35
35.1
35.2
35.3
RowNumber
RowCount28
36
36.1
36.2
36.3
RowNumber
RowCount29
37
37.1
37.2

3 4 5 6 7

35 >

RowNumber RowCount28

35.1 >

35.2 >

35.3 >

36 >

RowNumber RowCount29

36.1 >

36.2 >

36.3 >

37 >

RowNumber RowCount30

37.1 >

37.2 >

8 9

35 >

35.1 >

35.2 >

RowNumber RowCount21

35.3 >

36 >

36.1 >

36.2 >

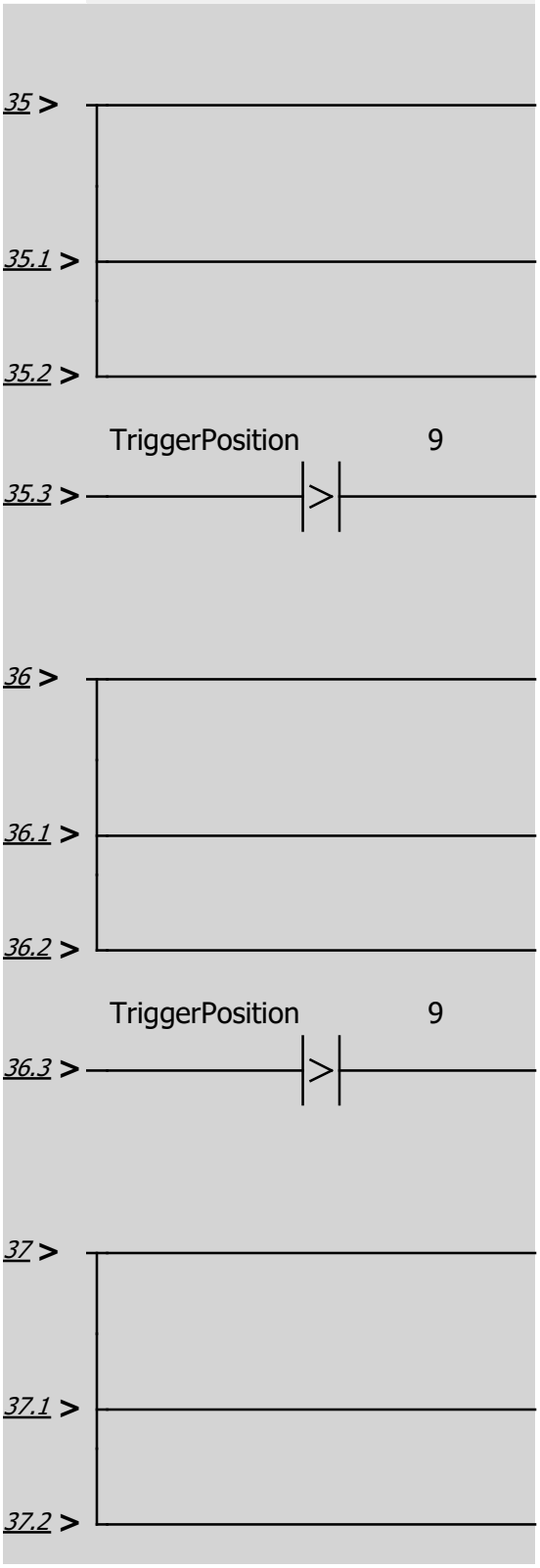
RowNumber RowCount21

36.3 >

37 >

37.1 >

37.2 >


35 >
35.1 >
35.2 >
TriggerPosition 9
35.3 >
36 >
36.1 >
36.2 >
TriggerPosition 9
36.3 >
37 >
37.1 >
37.2 >

11

35 > Enable
COPY DATA
Source Destination
HoleS28 DiameterIn
Type Copy Value

35.1 > Light21 ( SET )

35.2 > Light20 ( RST )

35.3 > Enable
COPY DATA
Source Destination
HoleS28 DiameterIn
Type Copy Value

36 > Enable
COPY DATA
Source Destination
HoleS29 DiameterIn
Type Copy Value

36.1 > Light21 ( SET )

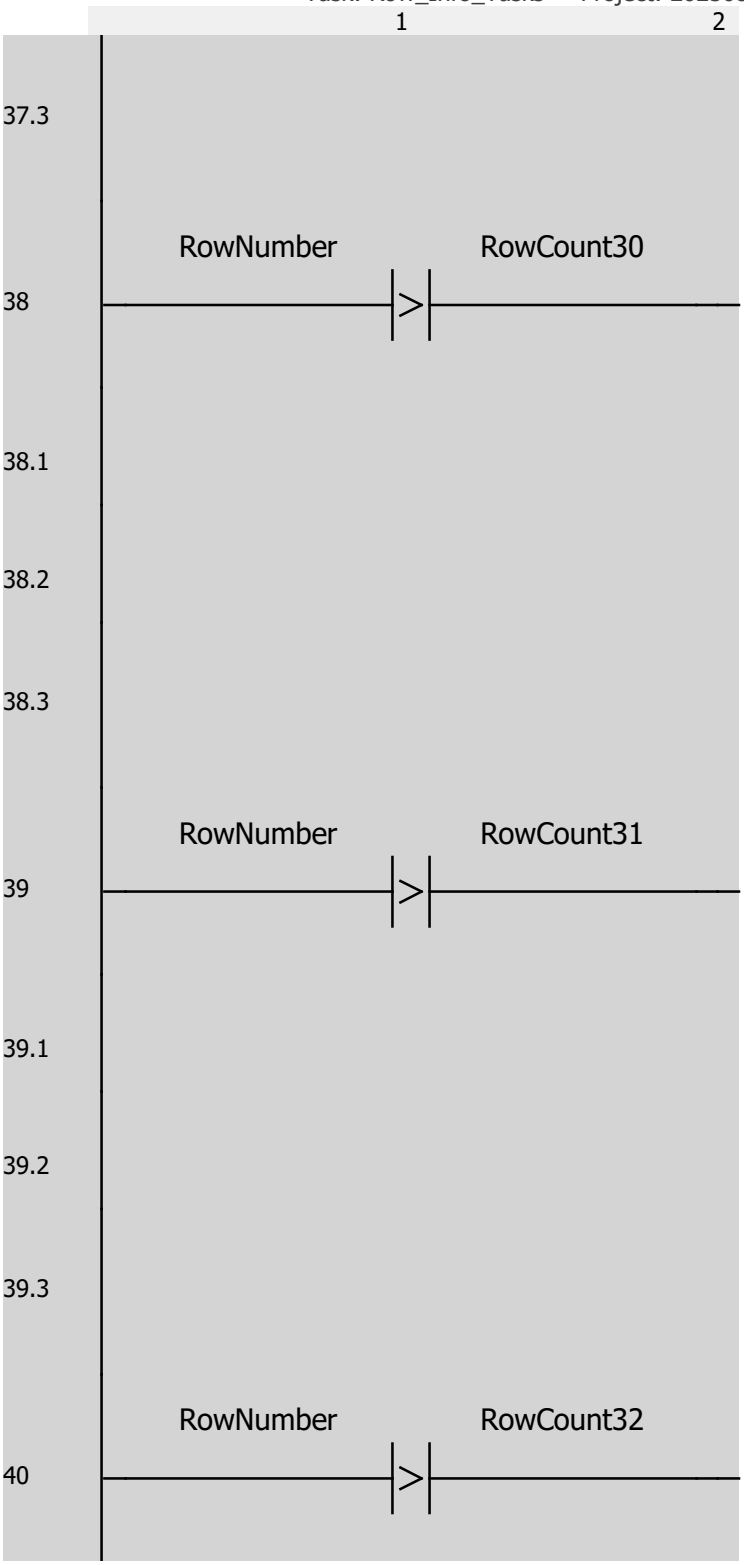
36.2 > Light20 ( RST )

36.3 > Enable
COPY DATA
Source Destination
HoleS29 DiameterIn
Type Copy Value

37 > Enable
COPY DATA
Source Destination
HoleS30 DiameterIn
Type Copy Value

37.1 > Light21 ( SET )

37.2 > Light20 ( RST )


1
2
37.3
RowNumber
RowCount30
38
38.1
38.2
38.3
RowNumber
RowCount31
39
39.1
39.2
39.3
RowNumber
RowCount32
40

3 4 5 6 7

37.3 >

RowNumber RowCount31

38 > ≤

38.1 >

38.2 >

38.3 >

RowNumber RowCount32

39 > ≤

39.1 >

39.2 >

39.3 >

RowNumber RowCount33

40 > ≤

8 9

37.3 >

RowNumber RowCount21

=

38 >

38.1 >

38.2 >

RowNumber RowCount21

38.3 >

=

39 >

39.1 >

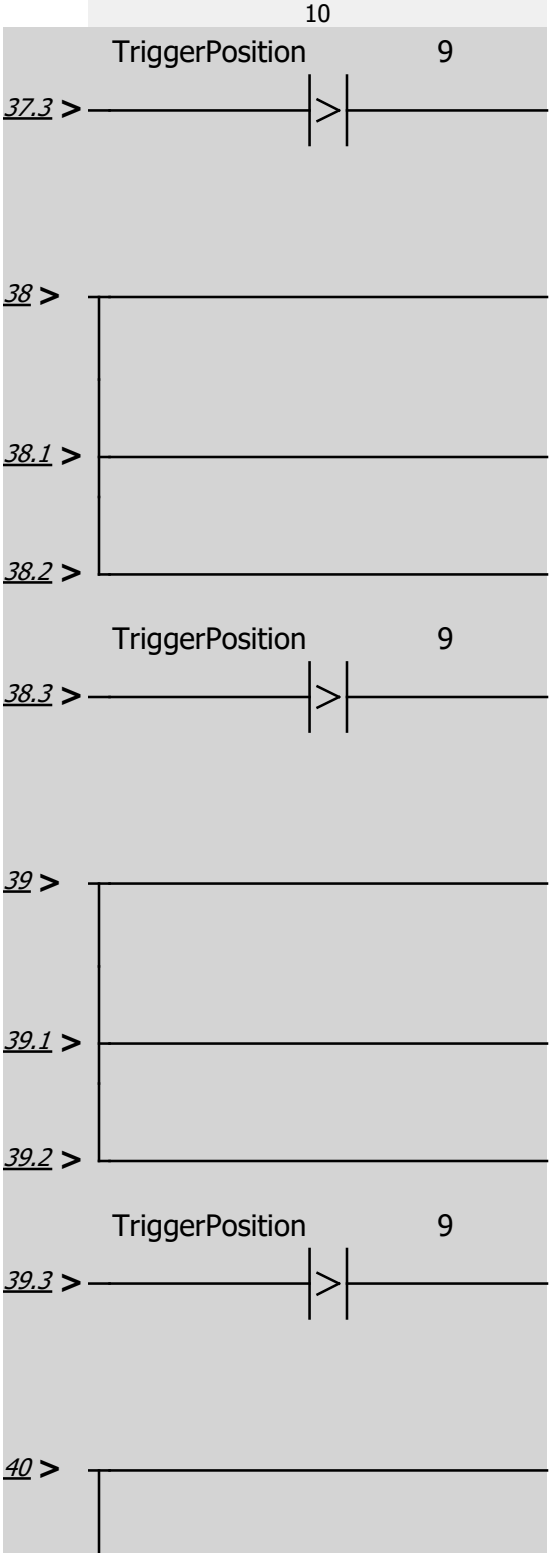
39.2 >

RowNumber RowCount21

39.3 >

=

40 >


TriggerPosition 9
37.3 >
38 >
38.1 >
38.2 >
TriggerPosition 9
38.3 >
39 >
39.1 >
39.2 >
TriggerPosition 9
39.3 >
40 >

11

37.3 > Enable
COPY DATA
Source Destination
HoleS30 DiameterIn
Type Copy Value

38 > Enable
COPY DATA
Source Destination
HoleS31 DiameterIn
Type Copy Value

Light21
38.1 > ( SET )

Light20
38.2 > ( RST )

38.3 > Enable
COPY DATA
Source Destination
HoleS31 DiameterIn
Type Copy Value

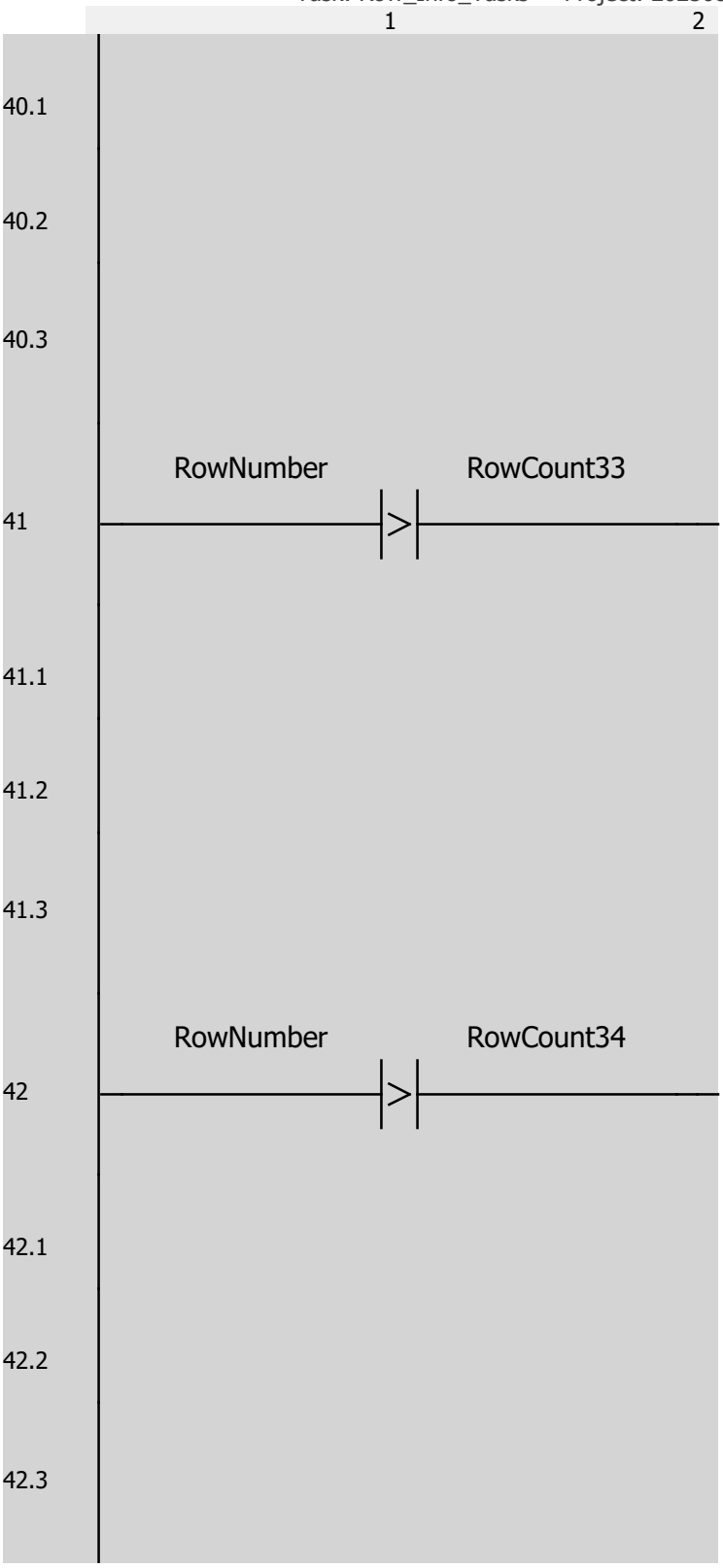
39 > Enable
COPY DATA
Source Destination
HoleS32 DiameterIn
Type Copy Value

Light21
39.1 > ( SET )

Light20
39.2 > ( RST )

39.3 > Enable
COPY DATA
Source Destination
HoleS32 DiameterIn
Type Copy Value

40 > Enable
COPY DATA
Source Destination
HoleS33 DiameterIn
Type Copy Value


1
2
40.1
40.2
40.3
RowNumber
RowCount33
41
41.1
41.2
41.3
RowNumber
RowCount34
42
42.1
42.2
42.3

3 4 5 6 7

40.1 >

40.2 >

40.3 >

RowNumber RowCount34

41 > ≤

41.1 >

41.2 >

41.3 >

RowNumber RowCount35

42 > ≤

42.1 >

42.2 >

42.3 >

8 9

40.1 >

40.2 >

RowNumber RowCount21

40.3 >

41 >

41.1 >

41.2 >

RowNumber RowCount21

41.3 >

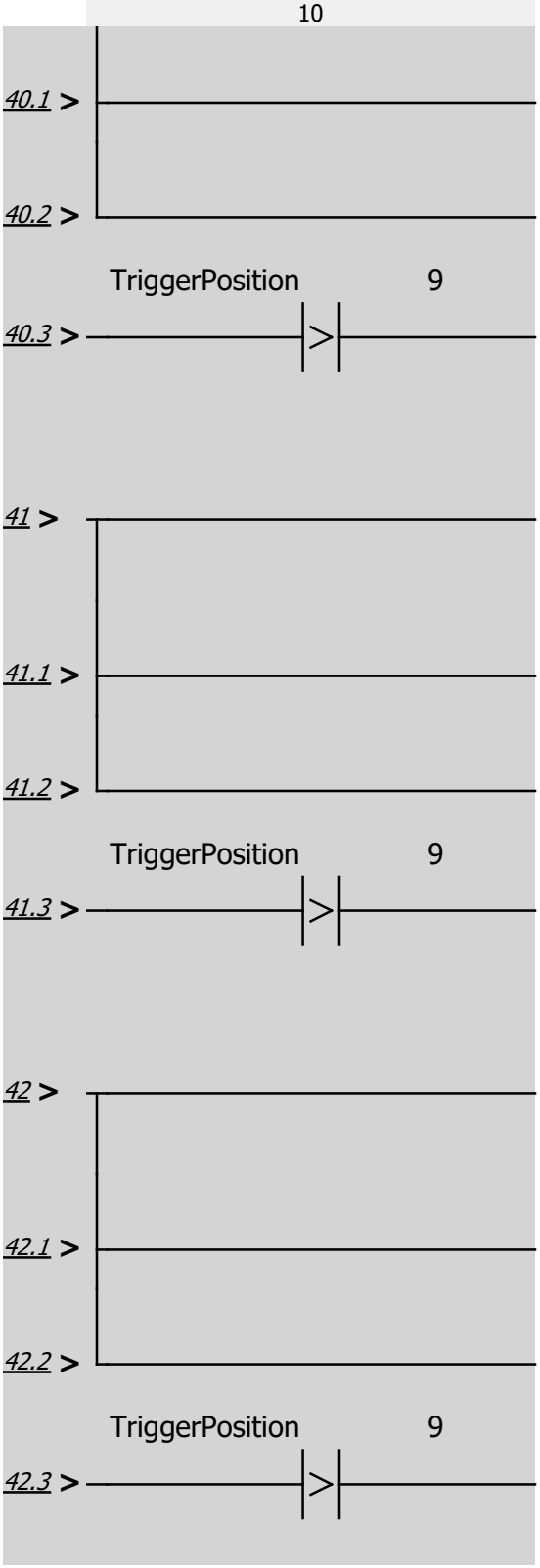
42 >

42.1 >

42.2 >

RowNumber RowCount21

42.3 >


40.1 >
40.2 >
TriggerPosition 9
40.3 >
41 >
41.1 >
41.2 >
TriggerPosition 9
41.3 >
42 >
42.1 >
42.2 >
TriggerPosition 9
42.3 >

11

Light21

40.1 > —( SET )

Light20

40.2 > —( RST )

40.3 > Enable

COPY DATA
Source Destination
HoleS33 DiameterIn
Type Copy Value

41 > Enable

COPY DATA
Source Destination
HoleS34 DiameterIn
Type Copy Value

Light21

41.1 > —( SET )

Light20

41.2 > —( RST )

41.3 > Enable

COPY DATA
Source Destination
HoleS34 DiameterIn
Type Copy Value

42 > Enable

COPY DATA
Source Destination
HoleS35 DiameterIn
Type Copy Value

Light21

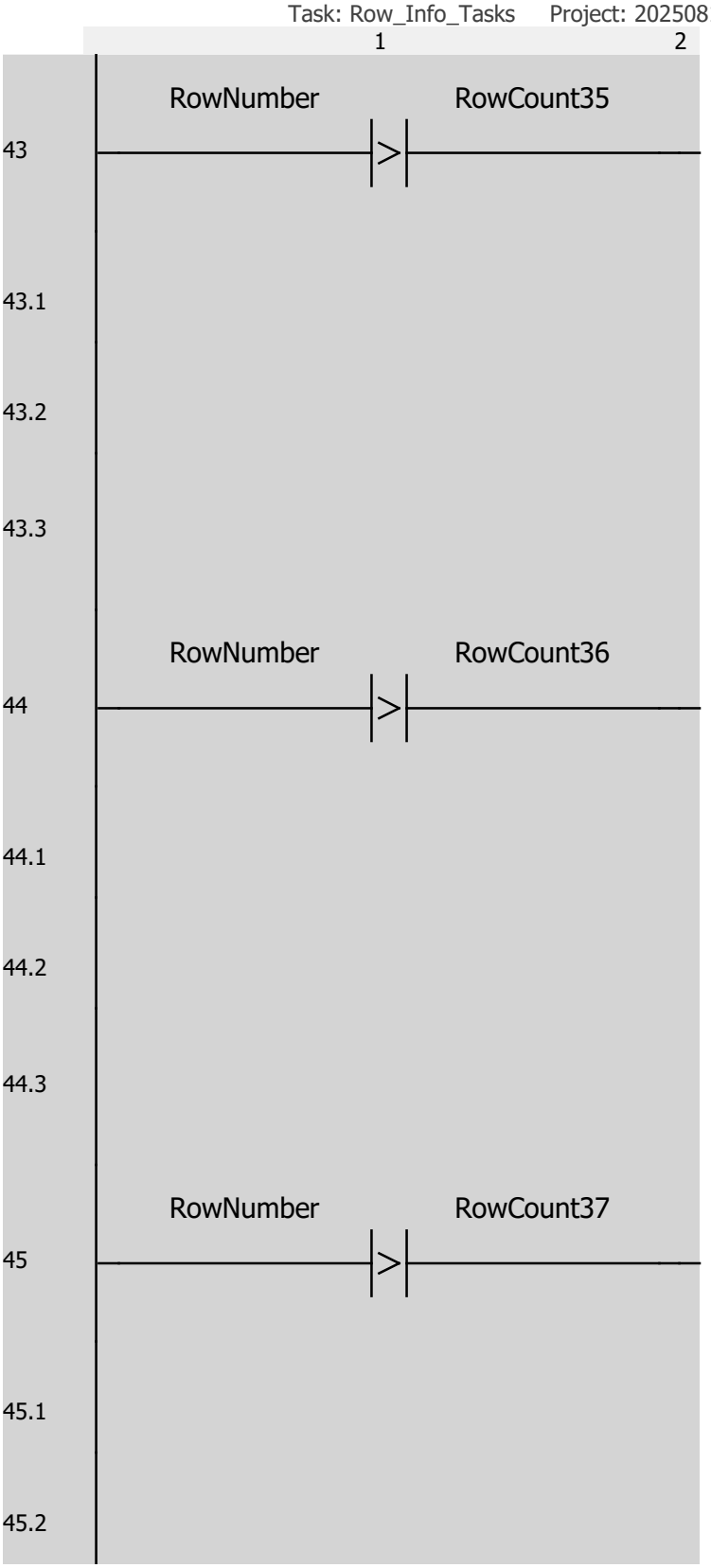
42.1 > —( SET )

Light20

42.2 > —( RST )

42.3 > Enable

COPY DATA
Source Destination
HoleS35 DiameterIn
Type Copy Value


1
2
RowNumber
RowCount35
43
43.1
43.2
43.3
RowNumber
RowCount36
44
44.1
44.2
44.3
RowNumber
RowCount37
45
45.1
45.2

3 4 5 6 7

43 >

RowNumber RowCount36

43.1 >

43.2 >

43.3 >

44 >

RowNumber RowCount37

44.1 >

44.2 >

44.3 >

45 >

RowNumber RowCount38

45.1 >

45.2 >

8 9

43 >

43.1 >

43.2 >

RowNumber RowCount21

43.3 >

44 >

44.1 >

44.2 >

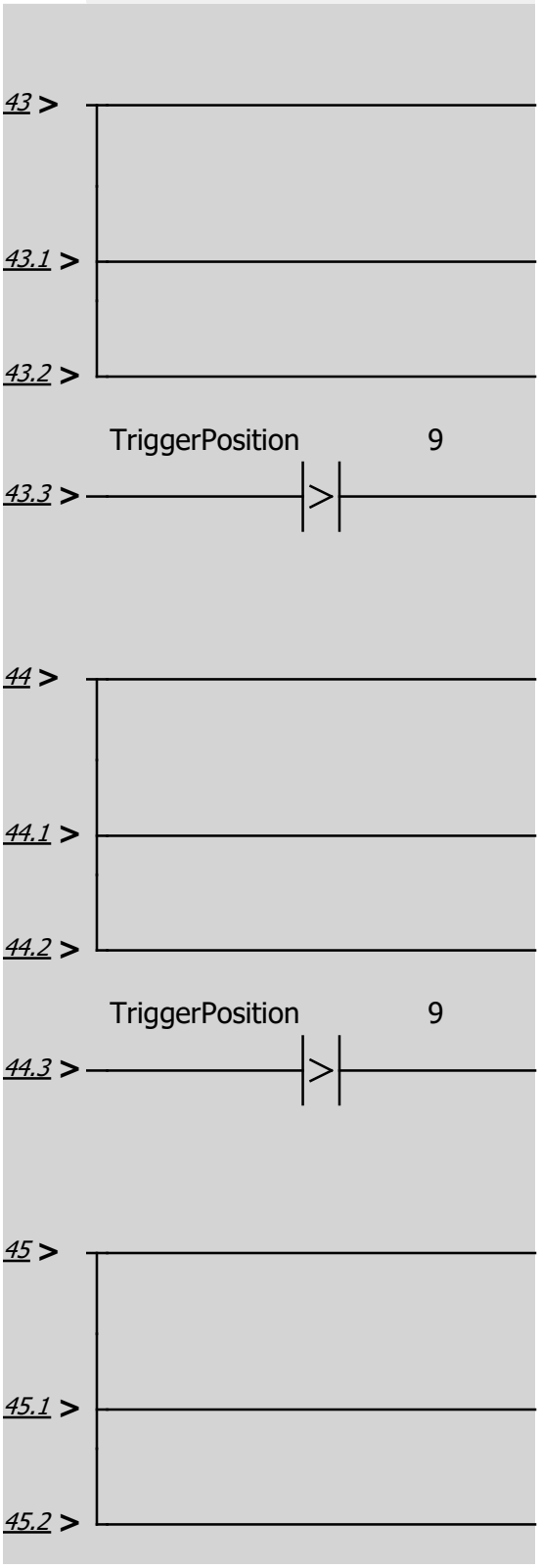
RowNumber RowCount21

44.3 >

45 >

45.1 >

45.2 >


43 >
43.1 >
43.2 >
TriggerPosition 9
43.3 >
44 >
44.1 >
44.2 >
TriggerPosition 9
44.3 >
45 >
45.1 >
45.2 >

11

43 > Enable COPY DATA
Source Destination
HoleS36 DiameterIn
Type Copy Value

43.1 > Light21 ( SET )

43.2 > Light20 ( RST )

43.3 > Enable COPY DATA
Source Destination
HoleS36 DiameterIn
Type Copy Value

44 > Enable COPY DATA
Source Destination
HoleS37 DiameterIn
Type Copy Value

44.1 > Light21 ( SET )

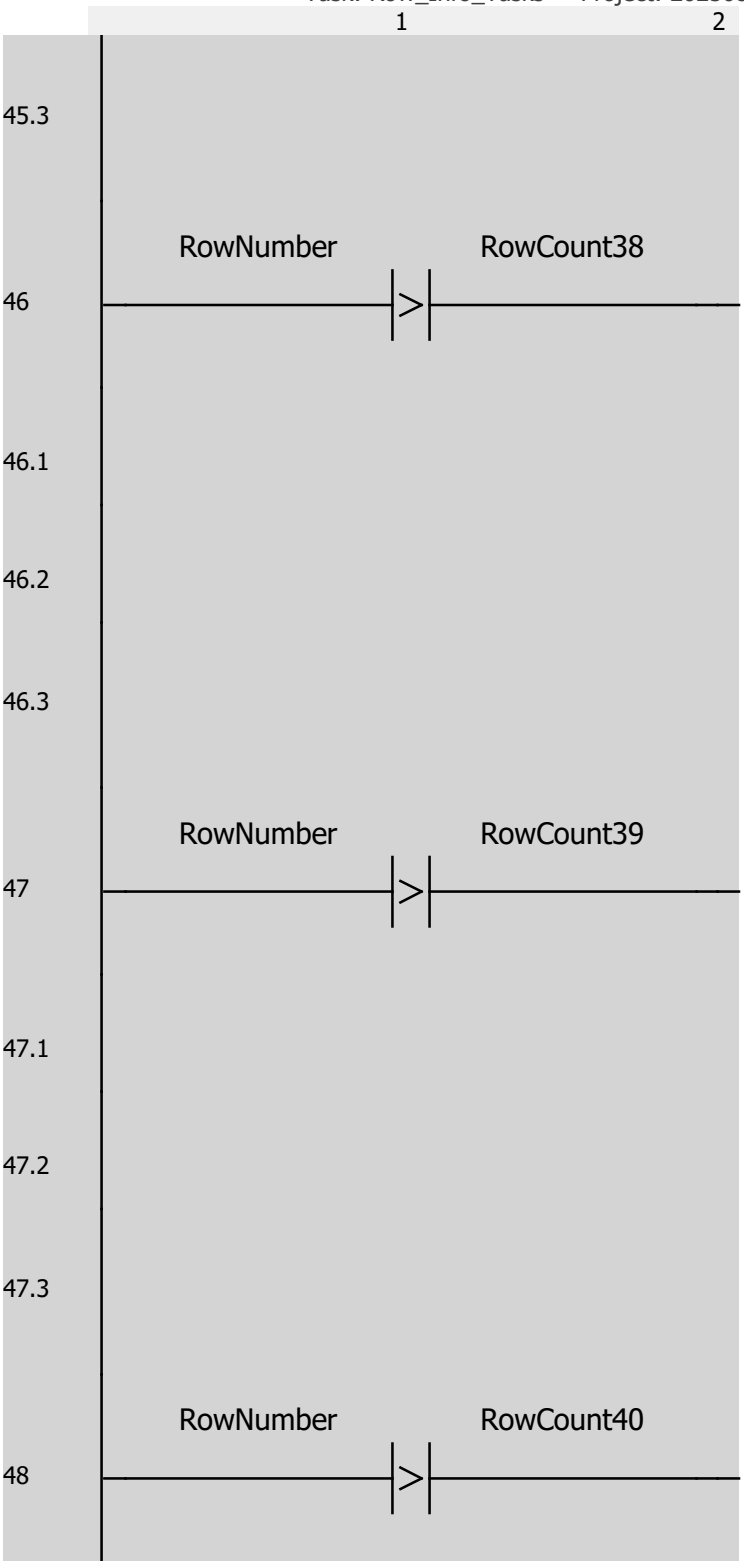
44.2 > Light20 ( RST )

44.3 > Enable COPY DATA
Source Destination
HoleS37 DiameterIn
Type Copy Value

45 > Enable COPY DATA
Source Destination
HoleS38 DiameterIn
Type Copy Value

45.1 > Light21 ( SET )

45.2 > Light20 ( RST )


1
2
45.3
RowNumber
RowCount38
46
46.1
46.2
46.3
RowNumber
RowCount39
47
47.1
47.2
47.3
RowNumber
RowCount40
48

3 4 5 6 7

45.3 >

46 > RowNumber ≤ RowCount39

46.1 >

46.2 >

46.3 >

47 > RowNumber ≤ RowCount40

47.1 >

47.2 >

47.3 >

48 > RowNumber ≤ RowCount41

8 9

45.3 >

RowNumber RowCount21

46 >

46.1 >

46.2 >

RowNumber RowCount21

46.3 >

47 >

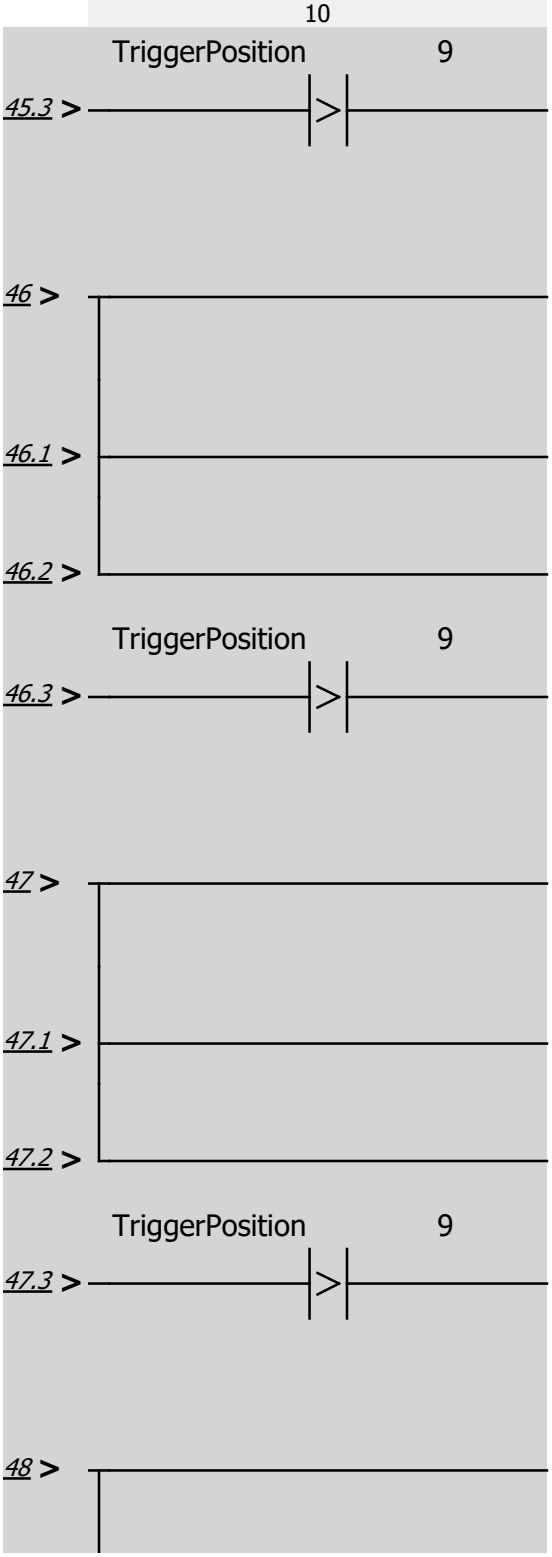
47.1 >

47.2 >

RowNumber RowCount21

47.3 >

48 >


TriggerPosition 9
45.3 >
46 >
46.1 >
46.2 >
TriggerPosition 9
46.3 >
47 >
47.1 >
47.2 >
TriggerPosition 9
47.3 >
48 >

11

45.3 > Enable
COPY DATA
Source Destination
HoleS38 DiameterIn
Type Copy Value

46 > Enable
COPY DATA
Source Destination
HoleS39 DiameterIn
Type Copy Value

Light21
46.1 > ( SET )

Light20
46.2 > ( RST )

46.3 > Enable
COPY DATA
Source Destination
HoleS39 DiameterIn
Type Copy Value

47 > Enable
COPY DATA
Source Destination
HoleS40 DiameterIn
Type Copy Value

Light21
47.1 > ( SET )

Light20
47.2 > ( RST )

47.3 > Enable
COPY DATA
Source Destination
HoleS40 DiameterIn
Type Copy Value

48 > Enable
COPY DATA
Source Destination
HoleS21 DiameterIn
Type Copy Value

1 2

48.1

48.2

48.3

49 RowNumber = RowCount13

50 RowNumber = RowCount5

51 RowNumber > 21

52 RowNumber > RowCount21

53 RowNumber > Total_Row_Number

54 RowNumber > RowCount6

55

56

3 4 5 6 7

48.1 >

48.2 >

48.3 >

49 >

PolyPipe.Ax1Position 9

50 > |>|

51 >

52 >

53 >

54 >

55 >

56 >

8 9

48.1 >

48.2 >

RowNumber RowCount21

48.3 >

49 >

50 >

51 >

52 >

53 >

54 >

55 >

56 >

10

48.1 >

48.2 >

TriggerPosition 9

48.3 > >

49 >

50 >

51 >

52 >

53 >

54 >

55 >

56 >

Light21

48.1 > —( SET )

Light20

48.2 > —( RST )

48.3 > Enable

| COPY DATA | |
|---|---|
| Source | Destination |
| HoleS21 | DiameterIn |
| Type | Copy Value |

Light13

49 > —( RST )

50 > Enable

| COPY DATA | |
|---|---|
| Source | Destination |
| 0 | DiameterIn |
| Type | Copy Value |

51 > Enable

| COPY DATA | |
|---|---|
| Source | Destination |
| 0 | DiameterIn |
| Type | Copy Value |

Light15

52 > —( RST )

53 > Enable

| COPY DATA | |
|---|---|
| Source | Destination |
| 0 | DiameterIn |
| Type | Copy Value |

Light6

54 > —( RST )

55 > —( END )

56 > —( END )

| | 1 | 2 |
|---|---|---|
| 57 | | |
| 58 | | |

| | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|
| 57 > | | | | | |
| 58 > | | | | | |

8 9

57 >

58 >

10

57 >

58 >

11

57 > ——————( END )

58 > ——————( END )

I/O Modules

| Part Number | Group Number | Base Number | Slot Number | Description |
|---|---|---|---|---|
| P2-550 | 0 | 1 | 0 | CPU Module |
| P2-06TRS | 0 | 1 | 1 | 6PT, ISOLATED OUTPUT RELAY |
| P2-04AD | 0 | 1 | 2 | 4CH, 16-BIT, VOLTAGE/CURRENT, ANALOG INPUT |
| P2-08SIM | 0 | 1 | 3 | 8PT, INPUT SIMULATOR |
| P2-SCM | 0 | 1 | 4 | Serial Communication Module |

GS Drives

| Part Number | Node Number | Horse Power Rating |
|---|---|---|

Protos X

| Module P/N | Group | Position | Module Description |
|---|---|---|---|

Modbus Devices

| Device Name | Port Type | Ethernet Port | IP Address Type | IP Address | Port No. | Ethernet Slave Node | Enable TCP Connect | Serial Port | Serial Slave Node | TCP Connecte | TCP/IP Error |
|---|---|---|---|---|---|---|---|---|---|---|---|

| Data Block | I/O T... | Enabl... | Tag Name | Data Type | Element Count | Message Size | Connection Point (I/O) | Class (Explicit) | Instance (Explicit) | Attribute (Explicit) |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Input | No | | | | | 101 (0x65) | 4 (0x4) | 101 (0x65) | 3 (0x3) |
| | Output | No | | | | | 102 (0x66) | 4 (0x4) | 102 (0x66) | 3 (0x3) |

Devices

| Device Name | Ethernet Port | IP Address | Port No. | Session Timeout Enable | Session Timeout in secs | Byte Order Swap | TCP Connected | Adapter Name | Vendor ID | TCP/IP Error |
|---|---|---|---|---|---|---|---|---|---|---|

Messages

| Device Name | Message Name | Enable Bit | RPI in msec | Routing Enabled | Slot No. | Connection Online | General Status | Extended Status | Status Description |
|---|---|---|---|---|---|---|---|---|---|

Data

| Device Name | Message Name | Data Name | Config Enabled | Delivery Option | RPI in msec | Connection Point | Data Array | Element Count (Msg Size) | Status Header Included |
|---|---|---|---|---|---|---|---|---|---|